

Received 18 July 2025, accepted 12 August 2025, date of publication 20 August 2025, date of current version 28 August 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3600984

 SURVEY

Adversarial Challenges in Network Intrusion Detection Systems: Research Insights and Future Prospects

SABRINE ENNAJI^{ID}, FABIO DE GASPARI^{ID}, (Member, IEEE), DORJAN HITAJ^{ID},
ALICIA KBIDI, AND LUIGI VINCENZO MANCINI^{ID}

Sapienza University of Rome, 00185 Rome, Italy

Corresponding author: Sabrina Ennaji (ennaji@di.uniroma1.it)

This work was supported in part by the PRIN 2022 20229BCXNW Project—Algorithmic Problems and Machine Learning funded by Italian Ministry of University and Research under Grant CUP B53D23012910006.

ABSTRACT Machine learning has brought significant advances in cybersecurity, particularly in the development of Intrusion Detection Systems (IDS). These improvements are mainly attributed to the ability of machine learning algorithms to identify complex relationships between features and effectively generalize to unseen data. Deep neural networks, in particular, contributed to this progress by enabling the analysis of large amounts of training data, significantly enhancing detection performance. However, machine learning models remain vulnerable to adversarial attacks, where carefully crafted input data can mislead the model into making incorrect predictions. While adversarial threats in unstructured data, such as images and text, have been extensively studied, their impact on structured data like network traffic is less explored. This survey aims to address this gap by providing a comprehensive review of machine learning-based Network Intrusion Detection Systems (NIDS) and analyzing their susceptibility to adversarial attacks, with a specific focus on evasion attacks due to their prevalence and real-world relevance. We critically examine existing research in NIDS, highlighting key trends, strengths, and limitations, while identifying areas that require further exploration. We also provide practical recommendations for adapting adversarial strategies to network-specific constraints. Additionally, we discuss emerging challenges in the field and offer insights for the development of more robust and resilient NIDS. In summary, this paper not only enhances the understanding of adversarial attacks and defenses in NIDS but also connects theoretical and practical perspectives, providing actionable guidance for future research to improve the robustness of machine learning models in cybersecurity applications.

INDEX TERMS Network intrusion detection systems, machine learning, cybersecurity, deep learning, adversarial attacks.

I. INTRODUCTION

The widespread deployment of internet-connected devices in our everyday lives has significantly impacted how we communicate, work, learn, and interact with the world. As society progresses in the information age, the Internet has evolved from a mere convenience to an indispensable necessity, playing a vital role in how our society functions,

businesses operate, and people connect. However, this high level of interconnectivity also creates opportunities for adversaries to exploit vulnerabilities in internet-facing services and devices for different purposes, including economic, political, or personal gain. Some attacks exploit network protocol vulnerabilities [1], [2], others focus on extracting and exploiting information through monitoring and sniffing network traffic [3], [4], while others merely use the network as a medium to reach and misuse vulnerable systems and services [5], [6].

The associate editor coordinating the review of this manuscript and approving it for publication was Wei Quan.

Given the threat that cyberattacks present, improving the security of networks and computer systems has been a growing concern. Modern defenses typically rely on defense-in-depth approaches involving a variety of different strategies, such as firewalls [7], cryptographic techniques [8], host-based malware detection [9], autonomous defense systems [10], [11], moving-target defense [12], [13], and Intrusion Detection Systems (IDS) [14]. Network Intrusion Detection Systems (NIDS), in particular, play a significant role in detecting various types of anomalous behavior due to their flexibility and ability to detect potential attacks before they reach a host by observing network traffic [15]. They are known to be efficient and have good potential to provide external and internal protection across services, devices, and architectures. In contrast to host-based IDS, NIDS do not need to run on each device that requires protection, but only require access to relevant network traffic. Despite these advantages, traditional NIDS based on expert knowledge require continuous updating to remain effective against evolving threats, as well as expert supervision to define detection rules. Over time, this can lead to performance degradation, reduced capability to detect novel attacks, and increased false positive rates. To address this gap, Machine Learning (ML) techniques and, more recently, Deep Learning (DL) have been increasingly integrated into NIDS to reduce the reliance on expert knowledge, improve generalization performance, and decrease the costs and complexity associated with updates [16]. ML, and particularly Deep Neural Network (DNN), has revolutionized several domains due to the ability to extract valuable features from data and make accurate predictions. This is demonstrated by its success in areas like image classification [17], [18], natural language processing [19], [20], speech recognition [21], [22], data generation [23], [24], cybersecurity [25], [26], forensics analysis [27], [28], and even applications related to the COVID-19 pandemic [29], [30].

Critically, while extremely effective in controlled environments, ML techniques are susceptible to various attacks when exposed to adversarial settings [31], [32]. Many attacks against ML have surfaced over the years, ranging from training-time attacks (e.g., backdooring, poisoning [33], [34]) to inference-time evasion attacks (e.g., adversarial examples [35], [36]). Evasion attacks, in particular, present a fundamental threat to NIDS, as attackers possess considerable flexibility in modifying their traffic patterns to evade detection. Adversarial examples fall in this category of attacks, manipulating legitimate traffic data with subtle modifications to deceive the ML model and achieve an undesired output (incorrect classification). While adversarial examples were originally designed for unstructured data in the continuous domain [35], the significant risk they create spurred research investigating their applicability to other domains, such as network traffic, in an effort to respond to the question:

Can adversarial attacks also threaten structured data?

Recent studies [37], [38], [39], [40], [41] demonstrate the applicability of evasion attacks and adversarial examples to structured data, introducing a significant vulnerability to cybersecurity applications, including ML-NIDS. However, existing research on NIDS design and evaluation often fails to adequately address this critical threat. This gap in knowledge motivates this work in three ways:

- **Raise Awareness:** The integration of adversarial attacks into the threat models of ML-NIDS remains limited in research. Many studies ignore these threats due to the increasing difficulties of adhering to feature interdependencies, semantic constraints, and real-world network traffic dynamics within structured data. Furthermore, adaptive attacks aimed to exploit these limitations and bypass detection measures are not adequately addressed in the context of NIDS. By addressing these major threats, this paper aims to raise awareness in the NIDS research community about the vulnerabilities of ML-based systems to adversarial threats and the necessity for proactive mitigation strategies.
- **Provide a Clear Point of Reference:** This developing field lacks comprehensive surveys on adversarial examples against NIDS. To the best of our knowledge, there is no existing survey that provides a full taxonomy of ML-NIDS with practical guidelines, explores the interplay between adversarial examples and feature interdependencies within NIDS, and focuses on the fundamental issues raised by evasion attacks. To fill this void, we aim to provide a new reference point for NIDS research, covering all aspects of robust network intrusion detection.
- **Establish Practical Guidelines:** It is challenging to compare ML-NIDS studies due to differences in datasets, metrics, and settings. By providing specific recommendations for practical implementation, this survey goes beyond theory and prioritizes adaptive attack testing, computational efficiency, feature interdependencies, and semantic validity.

A. CONTRIBUTIONS

This survey provides a comprehensive, analytical review of ML-based NIDS and recent practical applications of adversarial attacks against NIDS, considering emerging network threats, attacks, and defense strategies. In summary, we make the following contributions.

- 1) We review recent advancements in ML-based NIDS, providing a thorough taxonomy and analysis of existing approaches.
- 2) We compare and contrast different adversarial attacks against ML models, with a particular focus on evasion attacks due to their prevalence and practical importance. We reveal how these attacks exploit feature interdependencies and semantic restrictions within structured network traffic.

- 3) We discuss and critically review existing defenses against adversarial attacks on NIDS.
- 4) We provide an extensive analysis of lessons learned and novel insights, inviting researchers to investigate promising directions such as adversarial placement approaches, which could provide significant understanding of how attackers exploit network vulnerabilities. This contribution focuses on critical problems such as feature interdependencies, protocol conformance, and real-time applicability. We also present practical guidelines and best practices for reliable assessment and robust design of ML-based NIDS, with the goal of reducing the gap between theoretical research and practical deployment.

B. ORGANIZATION OF THE SURVEY

Figure 1 outlines a schematic representation of this paper's structure, which is organized as follows. Section II presents the methodology adopted to identify, select, and analyze the literature included in this survey. Section III compares this survey with existing IDS surveys, showing unique contributions such as the inclusion of a comprehensive taxonomy, practical recommendations, and exploration of adversarial placement strategies. Section IV provides general background on the evolution of NIDS, machine learning (ML), deep learning (DL), adversarial attacks, and their countermeasures. Section V presents a comparative overview of adversarial threats (poisoning, backdoor, and evasion attacks), analyzing their feasibility in NIDS, and explaining the rationale focus on evasion attacks. A comprehensive literature review of ML-NIDS is discussed in Section VI, providing foundational insights into the strong and key points of NIDS, which are further linked to their vulnerability to adversarial attacks that are analyzed in Section VII. The latter critically examines adversarial attacks on ML-NIDS showing how attackers exploit feature interdependencies and semantic constraints in network traffic. This has set the stage for Section VIII, which explores countermeasures. This section highlights the importance of proactive and adaptive defenses and links naturally to Section IX, which emphasizes adversarial placement as a critical consideration for designing stronger and more reliable defenses. Section X summarizes identified challenges, presenting lessons learned and guidelines. Finally, Section XI concludes our work and outlines potential avenues for future investigation in enhancing NIDS robustness.

II. METHODOLOGY AND ANALYTICAL DIMENSIONS

To ensure a comprehensive and reproducible review of adversarial attacks against ML-based NIDS, we adopted a structured and transparent methodological framework grounded in systematic review principles. Our primary objective was to identify, curate, and critically analyze advancements in adversarial attacks and defenses within this domain, with a focus on practical relevance, methodological validity, and existing research gaps.

A. RESEARCH QUESTIONS

Following an initial analysis of the literature and existing surveys in the field, our review was guided by these research questions:

- **Q1:** In current research on ML-based NIDS, which adversarial attacks are most frequently investigated?
- **Q2:** How do existing adversarial attack approaches consider the semantic and structural integrity of network traffic, and how do these factors affect their feasibility in realistic settings?
- **Q3:** How comprehensively do recent ML/DL-based IDS incorporate considerations of adversarial robustness, beyond optimizing for accuracy and performance?
- **Q4:** What defense strategies have been proposed and how effectively do they demonstrate practicality, adaptability, and generalizability across different adversarial scenarios and real-world deployment environments?

These questions shaped both our inclusion criteria and the analytical lens applied throughout the review.

B. LITERATURE COLLECTION AND SEARCH STRATEGY

We collected literature published between 2017 and 2024 from four major academic databases: **Google Scholar**, **IEEE Xplore**, **ScienceDirect**, and **ACM Digital Library**. These platforms were selected for their wide coverage and relevance to research on network intrusion detection, cybersecurity, and machine learning. We used Boolean term combinations to create our search queries, including:

```
("adversarial examples" OR "adversarial attacks" OR "evasion attacks")
AND ("machine learning" OR "deep learning")
AND ("network intrusion detection" OR "NIDS" OR "intrusion detection systems")
```

Search results were filtered to include full-text papers published in peer-reviewed leading journals and top-tier conferences. Due to the relatively limited number of peer-reviewed publications specifically addressing black-box evasion attacks on ML-based NIDS in the selected timeframe, we also included a small number of high-impact, highly cited preprints (e.g., [42], [43]) when they offered valuable technical insights or were frequently referenced in the literature.

Although the standardized terminology in this field made keyword-based queries generally practical, we manually screened titles and abstracts for contextual relevance, eliminated off-topic or duplicate results, and fine-tuned the final corpus using precise inclusion and exclusion criteria, as shown in Table 1 and discussed in detail through the following subsections. The reviewed literature remained focused, high-quality, and representative of the state of the art because of this multi-layered curation process.

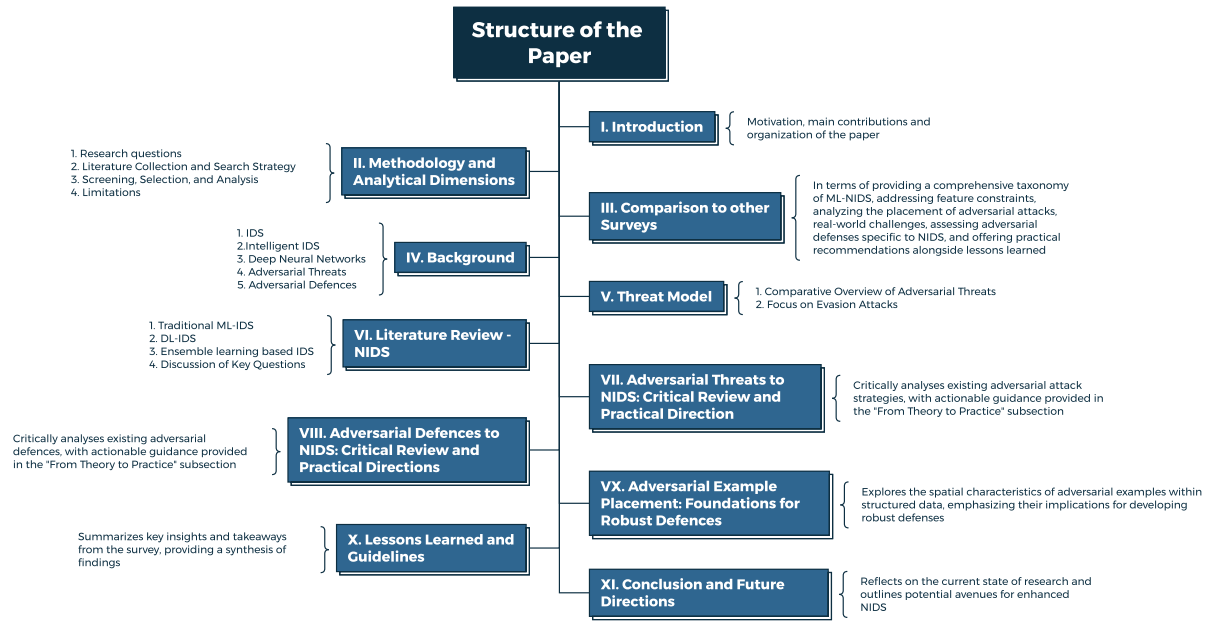


FIGURE 1. Structure of the presented survey.

C. SCREENING, SELECTION, AND ANALYSIS

A multi-step process was designed to ensure relevance, quality, and comprehensiveness in the selection and analysis of the literature.

• Phase I: Initial Screening

We initially retrieved over 220 papers based on keyword queries across Google Scholar, IEEE Xplore, ScienceDirect, and ACM Digital Library. Titles and abstracts were screened to exclude studies not pertinent to network intrusion detection or adversarial machine learning, resulting in a refined set of 142 candidate papers.

• Phase II: Full-Text Review

A comprehensive assessment of the 142 papers was conducted, evaluating their technical significance, incorporation of machine learning in NIDS, and empirical methodology. Papers that provided only theoretical insights without experimental evidence, or lacked focus on structured network traffic, were excluded. This process yielded a refined selection of 80 papers.

• Phase III: Eligibility Assessment

We then applied precise inclusion and exclusion criteria to maintain a consistent scope and ensure scientific rigor; a summary is provided in Table 1.

1) Inclusion Criteria: Papers were included if they

- addressed evasion attacks or defenses in ML-based NIDS
- provided detailed methodology, empirical results, or evaluation
- were published in reputable journals/conferences (e.g., IEEE Access, IJCNN, INFOCOM Workshops, Computers & Security)

- included widely cited preprints (e.g., arXiv) when peer-reviewed work was lacking, especially in black-box attack research
- proposed ML/DL-based NIDS improvements while ignoring adversarial robustness, revealing the lack of adversarial threat considerations within otherwise high-performing systems

2) Exclusion Criteria: Papers were excluded if they

- focused on non-network domains (e.g., computer vision, speech recognition)
- discussed traditional IDS approaches lacking ML integration
- lacked sufficient technical detail, evaluation, or methodological clarity

Based on the following analytical dimensions, we extracted and structured information for each selected study:

- **Attack types:** Categorized as white-box, grey-box black-box evasion attacks
- **Data constraints and realism:** Evaluated the extent to which proposed attacks respected the structural, semantic, and protocol-level constraints of real-world network traffic
- **Information gathering assumptions:** Assessed whether the research described the attacker's access model, probing techniques, or learning strategies; especially under black-box settings
- **State-of-the-art ML/DL-based NIDS approaches:** We analyzed a subset of recent ML/DL-based NIDS studies that focus on enhancing system performance (e.g., accuracy, scalability, feature selection). These were evaluated along axes such as architectural design, dataset usage, and vulnerability to adversarial examples,

TABLE 1. Summary of inclusion and exclusion criteria.

Included	Excluded
Studies on evasion attacks and corresponding defenses in ML-based NIDS	Studies on adversarial ML for non-network domains
Empirical studies with clear methodology and evaluation	Conceptual papers without experiments or concrete methods
Widely cited arXiv preprints on black-box setups	Low-quality or non-peer-reviewed and uncited grey literature
Publications from top journals/conferences	Articles in workshops or venues with unclear academic standards
Recent ML/DL-based NIDS papers that omit adversarial threats, included to highlight critical gaps	Traditional IDS approaches without ML components

even when adversarial robustness was not their primary focus (see Table 5). This helped identify recurring weaknesses (e.g., overfitting, sensitivity to noise, or misinterpretation of learned features) that could be exploited by adversarial attacks.

- **Defense techniques:** Classified by type (e.g., adversarial training, ensemble models, detection and rejection, etc.), and analyzed in terms of adaptiveness, generalizability, and real-world applicability.

In addition to ensuring that the final set of research fit within our defined scope, this integrated screening and analysis process allowed for a deeper understanding of the constraints and applicability of adversarial strategies in the context of NIDS.

D. LIMITATIONS

While our survey is, to the best of our knowledge, the first to provide practical recommendations grounded in a structured analysis of adversarial evasion attacks on ML-based NIDS, it is not without limitations. We restricted our review to the years 2017–2024 and only examined English-language papers that were listed in four significant academic databases. As a result, it's possible that some previous foundational work or relevant multidisciplinary research employing different language were missed. In addition, we excluded domains like endpoint protection, binary malware analysis, and behavioral anomaly detection; both to preserve thematic clarity and because adversarial robustness remains underexplored in those areas. Despite these limitations, our analysis provides a comprehensive picture of the current state of the field, identifies critical gaps, and offers actionable guidance to support the development of more robust and practical NIDS solutions.

III. COMPARISON TO OTHER SURVEYS

Adversarial attacks against NIDS is an emerging research area lying at the intersection of cybersecurity and ML. While extensive surveys exist on NIDS, there is a lack of comprehensive analysis specifically focused on adversarial attacks against them. Existing studies fall into three categories: (1) surveys on ML and DL-based NIDS, (2) surveys on adversarial attacks and defenses in the continuous domain, and (3) surveys on adversarial attacks on malware detection. In the following sections, we critically analyze related works,

identifying their limitations and highlighting the innovative aspects of our survey.

A. SURVEYS ON TRADITIONAL MACHINE LEARNING AND DEEP LEARNING APPLICATIONS TO NIDS

Several surveys analyze the application of traditional ML and DL to NIDS. A foundational work in this area is by Buczak et al. [44]. They conducted a comprehensive literature study on ML for network intrusion detection around 2016. Their work presents a complete taxonomy of existing NIDS solutions and provides tutorial descriptions of each ML applied therein. In addition to an overview and description of commonly used datasets to train and test NIDS. Building upon this foundation, Chaabouni et al. [45] analyzed ML-NIDS techniques for securing Internet of Things (IoT) devices. Similarly to Buczak et al., they propose a thorough taxonomy of NIDS, refining it for the IoT context to address domain-specific problems and challenges. Ahmad et al. [46] furthered this progress by defining an updated taxonomy of ML-NIDS and critically reviewing existing techniques, highlighting challenges and proposing new research directions. Many more general surveys on NIDS exist [47], [48], [49], providing similar contributions and including the most recent ML-NIDS approaches proposed in the literature. However, while these references are valuable for new researchers in the field, they all share a crucial limitation: they fail to consider the problem of adversarial attacks to ML-NIDS and related countermeasures.

B. SURVEYS ON ADVERSARIAL ATTACKS AND DEFENSES IN THE CONTINUOUS DOMAIN

Adversarial attacks against ML models in the continuous domain are a well-known challenge, both for traditional ML [31] and DNN [50], [51]. Barreno et al. [31] propose one of the first comprehensive analyses of the security of traditional ML techniques against adversarial attacks. In their work, the authors define a complete taxonomy of adversarial attacks against traditional ML algorithms and study the effectiveness of such attacks against practical applications such as spam filtering. In [50], Zhou et al. survey existing adversarial attacks against DNN, both at training and inference time. The authors provide a comprehensive classification for such attacks and define a lifecycle for adversarial attacks and defenses. Xiong et al. [51] propose a survey on the effects of different properties of training data

on the adversarial robustness of DNN. The authors analyze several published works and propose a set of guidelines and techniques for improving adversarial robustness through data manipulation.

This line of surveys provides an invaluable resource for understanding all the existing constructions for adversarial attacks and defenses. However, all of them focus on the continuous domain (mainly images) and are not applicable to the context of NIDS, where complex interdependencies exist between features and constraints on feature values. Moreover, they typically do not consider challenges specific to cybersecurity and NIDS.

C. SURVEYS ON ADVERSARIAL ATTACKS IN MALWARE DETECTION

1) GENERAL MALWARE DETECTION

Recently, researchers started studying the effects of adversarial examples in the area of malware detection. Gardiner et al. [52] studied the security of traditional ML techniques used for detecting malware's command and control channels. Their work systematically categorized existing command and control detection methods and analyzed the various attacks that could exploit these detectors. In [53], Park et al. provide a comprehensive survey of practical adversarial attacks against malware classifiers. Their paper explores various adversarial example-generation techniques, developing a framework to categorize them, and analyzes their strengths and weaknesses. More recently, Rosenberg et al. [54] propose in-depth survey on adversarial machine learning attacks and defenses specific to the cybersecurity field. Their study establishes a characterization of adversarial attacks based on three key aspects: lifecycle, adversarial goals, and attacker capabilities. Furthermore, they categorize different defense methods employed in cybersecurity applications.

These and other similar surveys (e.g., [55], [56]) respond to the broader challenges of adversarial attacks on ML and robust malware detection. However, they lack a comprehensive analysis specific to NIDS and often focus on a particular area in adversarial attacks, neglecting the unique issues that arise within NIDS. Finally, the analysis of potential defenses, when present, tends to be general and not specific to the context of malware detection.

2) INTRUSION DETECTION SYSTEMS

In [58], Olowononi et al. review ML-based defenses for Networked Cyber-Physical Systems (CPS), examining recent intrusion detection techniques in CPS that leverage ML, with a focus on their robustness against adversarial attacks. Their study provides a full taxonomy of traditional ML and DNN intrusion detection methods for CPS and identifies promising future research directions. While the authors investigate the challenge of developing robust ML applications for intrusion detection, their focus lies in the distinct domain of CPS, separate from the area addressed by our

survey. Our paper provides a comprehensive taxonomy and in-depth analysis of ML-NIDS, identifying and exploring the unique challenges within this domain. Additionally, it explores methods specifically designed for adversarial attacks against NIDS, in contrast to [58], which focuses on general adversarial attacks. In [59], Martins et al. survey applications of adversarial machine learning in intrusion and malware detection; they categorize general attack strategies and defenses before applying them to intrusion detection. Notably, their study lacks a general taxonomy of NIDS. Our survey distinguishes itself by presenting a detailed categorization of both ML-NIDS and the specific adversarial attacks that target them. We further present a thorough analysis of defenses against adversarial attacks for NIDS, which is an absent aspect in [59]. Moreover, we discuss new insights, valuable lessons learned, and promising research directions in this field. While in another thorough summary of several adversarial attacks and defenses for ML-NIDS explored by Ibitoye et al. [57], it does not provide a specific taxonomy of NIDS, but covers a wide variety of detectors such as spam detection, phishing detection, and general malware detection. Moreover, it does not consider the applicability of the discussed techniques. The presented defenses are also general and not specific to NIDS. Additionally, in [60] an analysis of adversarial attacks with related defenses for ML-IDS is provided, but without discussing existing gaps in this area. A comprehensive taxonomy of ML-NIDS is not considered as well. Similarly, Pinto et al. [61] survey applications of ML to intrusion detection to protect critical infrastructure. A high-level categorization of ML-based defenses is introduced, but the challenge of adversarial examples is not thoroughly addressed. Finally, He et al. [62] propose a comprehensive analysis of existing works in ML-NIDS. They present a taxonomy of NIDS, a categorization of adversarial attacks and defenses against NIDS. In addition to an analysis of future research directions for this field. Compared to [62], not only does our survey include additional recent papers in the literature, but we also critically review existing techniques, highlighting limitations and potential challenges to adoption. Moreover, we discuss new observations and correlations between the placement of adversarial examples within the data manifold [63] and the resilience to adversarial attacks.

D. COMPARISON SUMMARY

In summary, compared to related works, our survey is distinguished by its comprehensive and methodical approach. We provide a complete taxonomy of ML-based NIDS, adversarial attacks targeting them, and their corresponding defenses. In contrast to existing surveys, which often focus on listing approaches, we critically evaluate their strengths and limitations, providing practitioners with actionable insights and practical recommendations.

Additionally, we explore the real-world applicability of these attacks with their defenses in the NIDS context. Notably, our survey shed light on new discoveries about

TABLE 2. Contributions of this survey in comparison to the most relevant existing works.

Reference	Year	ML-NIDS Taxonomy	Feature Interdependency	Placement of Adversarial Attacks	Real-World Challenges	NIDS Adversarial Defenses	Practical Recommendations	Lessons learned
Ibitoye et al. [57]	2019	✗	✓	✗	✗	✓	✗	✓
Olowononi et al. [58]	2020	✓	✗	✗	✗	✗	✗	✗
Martins et al. [59]	2020	✗	✓	✗	✓	✗	✗	✗
Alotaibi et al. [60]	2023	✗	✗	✗	✓	✗	✗	✗
Pinto et al. [61]	2023	✗	✓	✗	✓	✗	✗	✗
He et al. [62]	2023	✓	✓	✗	✗	✓	✗	✓
Our survey	2024	✓	✓	✓	✓	✓	✓	✓

the existence and placement of adversarial examples, which provides a novel perspective on how the spatial positioning of adversarial examples can influence defense mechanisms. We further investigate the interdependencies between malware features in NIDS, uncovering important insights for developing robust models. By analyzing these aspects and conducting a comprehensive review, we extract valuable lessons learned and propose a set of guidelines for developing robust ML-NIDS. Table 2 summarizes the comparison of our survey to the most relevant related works and highlights significant gaps in existing surveys on adversarial examples against ML-NIDS.

Surveys serve an indispensable purpose in consolidating knowledge and identifying research directions, which supports the advancement of a discipline. We believe our work offers detailed analysis and novel insights, making it a valuable resource for both new and experienced NIDS practitioners.

IV. BACKGROUND

This section provides background information on intrusion detection systems, deep neural networks, common adversarial attacks on learning systems, and existing defenses.

A. INTRUSION DETECTION SYSTEMS

Intrusion Detection Systems include network traffic monitoring (NIDS) and system behavior analysis (host-based IDS) to detect cyberattacks [64]. They work in conjunction with other security mechanisms, including firewalls, and use various detection techniques categorized into two main types:

- **Signature-based detection** uses information collected from previous attacks and known vulnerabilities to create a characteristic and recognizable signature. As attackers continuously update their attack patterns and methods, signatures become quickly obsolete and need to be regularly updated to remain effective [65]. Given its reliance on known attack patterns, signature-based detection typically offers high precision with low false positive rates but fails to detect new, unknown attacks.
- **Anomaly-based detection** models the typical behavior of systems and networks and detects deviations from such behavior as potential attacks. The main advantage of this approach lies in its ability to identify both known and unknown attacks, as long as they cause a detectable

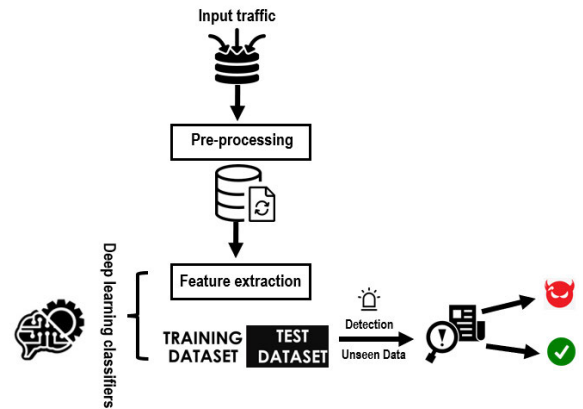


FIGURE 2. High-level architecture of a deep learning based-IDS.

behavioral shift. Furthermore, once the typical behavior of a system/network is computed, anomaly-based IDS do not need continuous updates. However, anomaly-based detection generally results in a higher rate of false positives compared to signature-based methods, making it more time-consuming to handle false alerts.

B. INTELLIGENT INTRUSION DETECTION SYSTEMS

The growing complexity of cyberattacks led to significant advances in IDS. In recent years, there has been a significant push towards the integration of ML techniques into IDS, greatly enhancing detection capabilities [66]. A growing body of research highlights that ML mechanisms consistently outperform traditional IDS methods in various settings, with DL algorithms, in particular, showing promising results [64]. This advantage arises from the ability of DL models to automatically learn and select descriptive features, eliminating the dependency on complex, expert-driven feature engineering (see Figure 2); a major limitation in traditional ML [67]. Additionally, DL models excel at handling high-dimensional data and identifying intricate relationships between features, resulting in improved classification accuracy [68]. They also demonstrate strong adaptability to the dynamic nature of network traffic and evolving attack patterns. Although DL training can be computationally expensive, advances in software and hardware are mitigating this challenge [69]. Despite their numerous advantages, both ML- and DL-NIDS share a significant vulnerability: **adversarial attacks** [37], [70]. Adversarial attacks involve adversarial perturbation of training or query data designed to cause the model to

fail in detecting malicious activity, thereby undermining its effectiveness.

C. DEEP NEURAL NETWORKS

Deep neural networks (DNNs) are algorithms developed to autonomously recognize, extract, and learn key patterns and relationships from input data, eliminating the need for extensive feature engineering by domain experts. DNNs perform well at processing large volumes of high-dimensional data, and have consistently redefined state-of-the-art performance across various learning tasks. Practically, a DNN can be represented as a function (or a set of functions) that given an input provides an output as follows:

$$f \in F : \mathbb{R}^n \rightarrow \mathbb{R}^m \quad (1)$$

Deep learning system comprises of two stages, the training and inference stages.

- **The training stage:** Let X be the class of inputs and Y a label space. The learning procedure for a DNN model f consists of the selection of a set of parameters θ that minimize a loss function l on the training set $\mathcal{D} = (\mathcal{X}, \mathcal{Y})$ which measures how well the DNN has learned over the data. The learning the parameters θ occurs through an iterative process where the loss function l , which measures the performance of the learned function f , provides the necessary information to optimize the parameters until an optimum is reached. More formally, neural network training seeks to optimize the following problem:

$$\underset{\theta}{\operatorname{argmin}} \sum_{(x,y) \sim \mathcal{D}} l(\theta, x, y) \quad (2)$$

where θ are the parameters of the DNN to optimize, x and y are the training datapoints and their corresponding labels sampled from the training data $\mathcal{D}$, and l is the loss function used for training.

- **The inference stage:** After finding an optimal set of parameters θ , the model can be used to classify new data. An unseen input x_j and its corresponding label y_j are calculated as follows:

$$y_j = f(\theta, x_j) \quad (3)$$

Additionally, during the classification task, the output of the last layer Y (called *logits*) are transformed into a probability distribution by an activation function (e.g., Softmax):

$$0 \leq Y_i \leq 1 \wedge Y_1 + \dots + Y_m = 1;$$

where Y_i can be understood as the probability that input x has class i . The label that has the highest probability $\operatorname{argmax}(Y_i)$ presents the class of the input x .

D. ADVERSARIAL THREATS

Traditional ML and DL techniques have been show to be vulnerable to adversarial attacks: manipulations of input data that lead models to produce wrong predictions. One

particularly concerning form of such attacks are *adversarial examples* [35], [71]. Initially studied in image classification task, adversarial examples involve subtle, often imperceptible perturbations to the input data that can significantly alter DNN's output.

Generally, the training process of a DNN involves minimizing a loss function l , which quantifies the error between the target and the predicted classes. This optimization process aims to reduce the error rate, improve the model's accuracy by adjusting its parameters based on the computed loss. The overall error rate used to evaluate the model's performance is then calculated by comparing the sum of the loss between the target and the corresponding labels.

$$\sum_{(x,y) \sim \mathcal{D}} l(f(\theta, x), y) \quad (4)$$

Adversarial examples are crafted by adding a small *adversarial perturbation* to a clean input x . The details on how the perturbation is generated depend on the specific algorithm $\mathcal{A}$ used, but for DNNs, they typically rely on gradient computation to increase the loss (i.e., error) of the model on a specific input x :

$$x_{adv} = (x + \mathcal{A}(\theta, x, y)) \quad (5)$$

Over the years, numerous adversarial attacks have been introduced, and they can be categorized based on four main criteria:

- 1) **Adversary's Knowledge:** Adversary capabilities are classified by the target model's access level.
 - **White box-attacks:** Full access to the model's architecture, training data, parameters, allowing highly effective adversarial examples. Common methods include Limited-memory Broyden–Fletcher–Goldfarb–Shanno [71], Fast Gradient Sign Method [35], Basic Iterative Method, Projected Gradient Descent [72], Deepfool [73], Carlini and Wagner [74], and Jacobian-based Saliency Map Attack [75].
 - **Black-box attacks:** No access to the model's internals, and can only interact with it in an oracle-like manner by querying specific inputs and observing the corresponding outputs. Consequently, black-box attacks typically rely on iterative methods, where the adversarial sample is updated over time to achieve the desired adversarial goal. These attacks often employ techniques like Generative Adversarial Networks (GAN) [59], [76] or Zeroth-Order Optimization attacks [77], but can also utilize on surrogate models by approximating the target model's behavior and then use these learned models to generate adversarial inputs for the original system [78].
 - **Gray-box attacks:** Partial knowledge about the target model. Unlike black-box access, where only the inputs and outputs of the model are observable, gray-box access allows some limited understanding

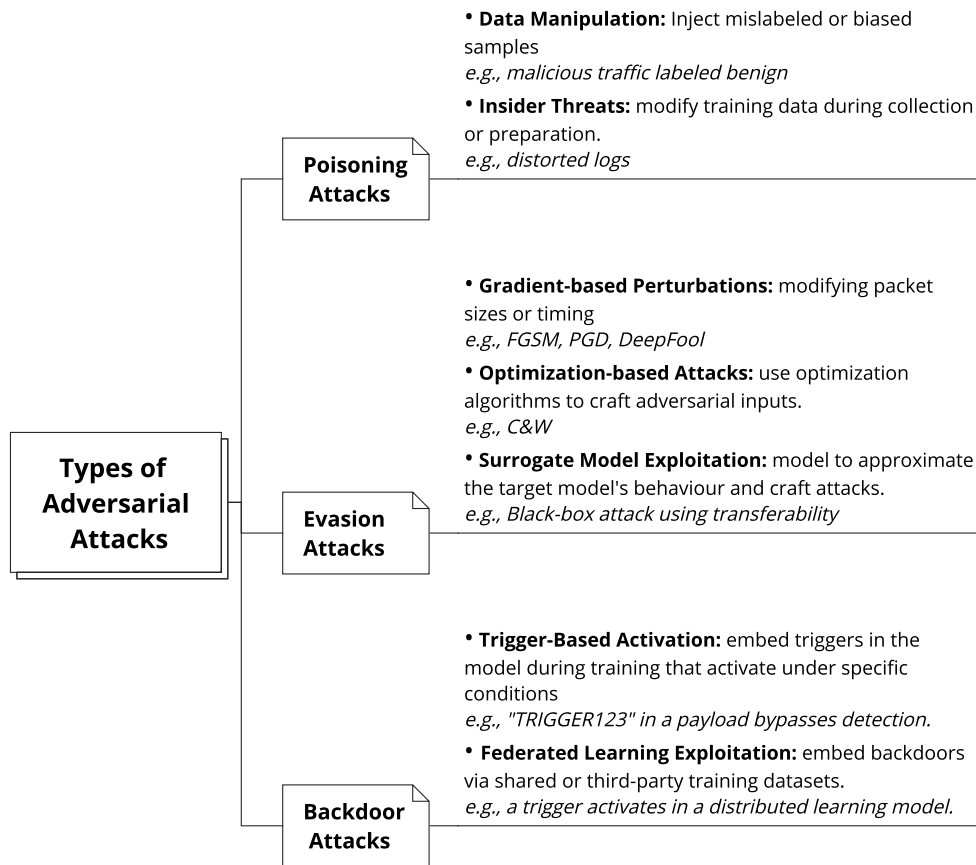


FIGURE 3. Categories of adversarial attacks.

or control over the model's internal structure or behavior. This can include access to certain layers or parameters but not full transparency like white-box access.

2) **Attack Timing:** Attacks are categorized according to the targeted phase of the machine learning process; training or inference. Each phase presents unique vulnerabilities that adversaries can exploit, as illustrated in Figure 3.

- **Poisoning Attacks (Training Phase):** They are introduced during the training phase, where attackers inject perturbed or malicious data into the training set. They aim to corrupt the model's learning process in order to cause misclassification during inference [33].
- **Evasion Attacks (Inference Phase):** They occur during the test phase, where attackers manipulate input samples in order to evade detection. The purpose of these attacks is to trick the model into making wrong predictions without changing the input's functionality [59], [79], [80].
- **Backdoor Attacks (Training and Inference Phases):** They insert hidden triggers into the model during training, which cause malicious behaviour when certain conditions are satisfied during inference. These triggers are intentionally created to stay inactive

unless a particular feature combination or input pattern is present. For example, a trigger generate when a particular set of protocol fields and packet sizes (e.g., "protocol_type" = UDP, "src_port" = 12345, "dst_port" = 54321) occur within network traffic, leading the model to incorrectly classify malicious traffic as benign [81].

- 3) **Adversarial Goal:** Adversarial attacks can be either targeted or untargeted. Targeted attacks aim to misclassify the input as a specific incorrect class. Untargeted attacks aim to force the model to misclassify the input, without specifying which class it should be classified.
- 4) **Attack Frequency:** Adversarial attacks can be one-shot attacks or iterative attacks. One-shot attacks generate adversarial perturbations are through a single-step algorithm. Iterative adversarial attacks rely on multiple iterations, repeatedly querying the classifier to refine the adversarial perturbation.

E. ADVERSARIAL DEFENSES

Adversarial examples have revealed vulnerabilities in ML and DL, including in NIDS. Over time, researchers have proposed defenses like adversarial training, model regularization, and input preprocessing to improve robustness.

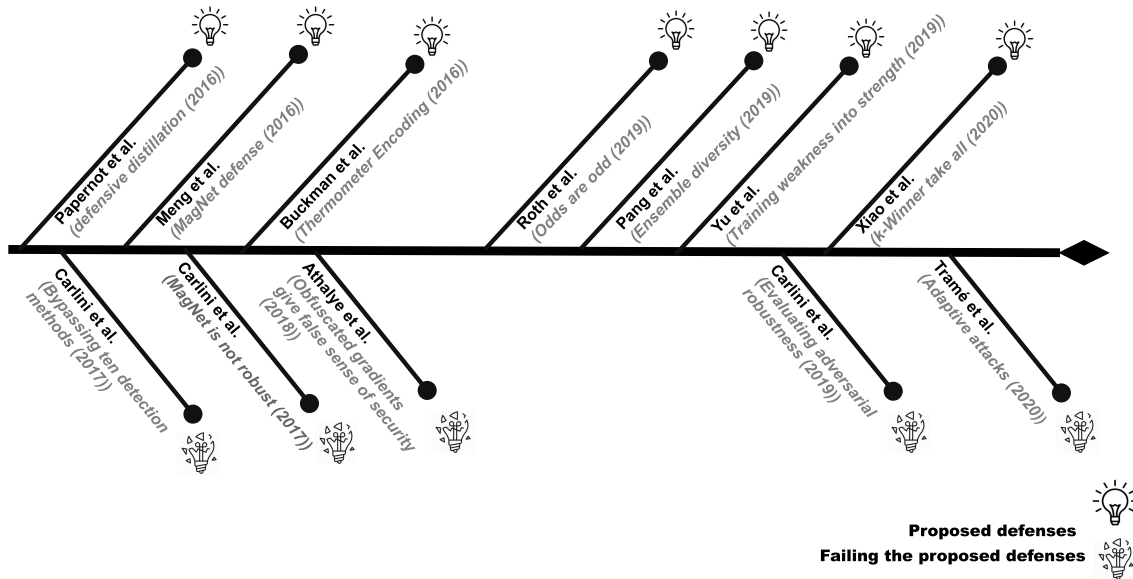


FIGURE 4. Timeline of adversarial attacks and defenses.

However, many fail against adaptive attacks, which evolve to evade them. Figure 4 illustrates the evolution of defenses and corresponding adaptive attacks [82], [83], [84], [85], [86].

This section categorizes existing defenses into two groups: (1) Adversarial training, which aims to anticipate attacks during model development, and (2) architecture reinforcing, which makes the models by nature resilient to adversarial perturbations.

- 1) **Adversarial Training:** It aims to enable the model to develop a robust representation of the training data, thereby reducing the impact of adversarial examples. This is accomplished by training the model using both clean and adversarial datapoints, allowing for an adaptation of the learned decision boundary to accurately classify both perturbed and unperturbed instances [35]. Formalizing, adversarial training aims to find a set of model parameters θ that minimize the following expression:

$$\underset{\theta}{\operatorname{argmin}} E_{(x,y) \sim D} [\max_{\delta \in \Delta} l(\theta, x + \delta, y)] \quad (6)$$

where δ is an adversarial perturbation, Δ represents the set of all allowable adversarial perturbations, D denotes the data distribution over inputs $x \in \mathbb{R}^d$ and labels y , $l(\theta, x, y)$ is the loss function for the model parameters θ .

- 2) **Architecture Reinforcing:** It includes several methods that share the common purpose of reinforcing the model's architecture, each employing different strategies, as detailed below.

- **Detection and Rejection:** It involves using trained detector $g(x)$ to evaluate model inputs and detect potential adversarial perturbations. When a perturbation is detected, the system executes a series of specialized operations to reduce/eliminate the identified threat [87]. Inputs exceeding a predefined threshold T

are flagged and either rejected or transformed:

$$g(x) = \mathbb{I} (\|f_{\theta}(x) - f_{\theta}(x_{\text{clean}})\| > T) \quad (7)$$

where $f_{\theta}(x)$ is the model's output for input x , x_{clean} is the reconstructed clean input, and $\mathbb{I}$ is the indicator function.

- **Input Transformation:** It applies a set of transformations to the input data to eliminate potential adversarial perturbations before the sample is used to query the model. The input transformations typically aim at disrupting the features altered by adversarial perturbations, mitigating the effect of the attacks [88]. Let $T(x)$ describes the transformation function applied to the input x :

$$x_{\text{transformed}} = T(x), \quad (8)$$

where $T(x)$ can include:

- **Noise reduction:** $T(x) = \text{filter}(x)$, which removes high-frequency components likely produced by adversarial perturbations.
- **Feature smoothing:** $T(x) = x - \nabla_x \mathcal{L}(f_{\theta}(x), y)$, which minimizes the gradient contributions of perturbed features to stabilize the input.

By applying these transformations, the effectiveness of adversarial examples against the model can be reduced by neutralizing or significantly weakening them.

- **Gradient Masking:** It aims to obscure/modify the gradient information of the model to limit adversarial manipulations. Most adversarial attacks rely on computing gradients to identify optimal perturbations. The model is less susceptible to these perturbations if this gradient information is restricted

or distorted [89]. It can also be expressed as:

$$\tilde{\nabla}_x \mathcal{L}(f_\theta(x), y) \approx 0, \quad (9)$$

where $\tilde{\nabla}_x$ is the masked or approximated gradient, and $\mathcal{L}(f_\theta(x), y)$ is the loss function computed for the model f_θ , input x , and label y .

This approach ensures that adversarial perturbations δ , typically computed as:

$$\delta = \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}(f_\theta(x), y)), \quad (10)$$

cannot be effectively produced because the true gradient $\nabla_x \mathcal{L}$ is unavailable/unreliable. By generating randomness or intentionally modifying the gradient, models can resist gradient-based attacks such as FGSM and PGD, albeit sometimes at the cost of low accuracy.

- **Manifold Projection:** It leverages the natural data distribution (manifold) to detect and remove adversarial perturbations [63], [90]. Adversarial examples typically reside in different region of the domain compared to clean samples, and manifold projection can be employed to discard adversarial examples that deviate significantly from the natural data structure. The process can be presented as:

$$x_{\text{projected}} = \underset{z \in \mathcal{M}}{\text{argmin}} \|x - z\|_p, \quad (11)$$

where $\mathcal{M}$ presents the manifold of clean data, $\|x - z\|_p$ is the p -norm (commonly $p = 2$, Euclidean distance) measuring the distance between the input x and the nearest point z on the manifold.

By reducing the deviation from the natural structure of clean data, the manifold projection ensures that the perturbed input x is returned to a valid data distribution. The manifold $\mathcal{M}$ can be learned using techniques like autoencoders or generative models [91]. Given an autoencoder with an encoder E and a decoder D , the manifold projection can be represented as:

$$x_{\text{projected}} \approx D(E(x)), \quad (12)$$

where $E(x)$ maps the input x to a lower-dimensional representation, and $D(E(x))$ reconstructs x from its encoded representation.

By mapping adversarial examples back to the natural data space, this method disturbs aggressive perturbations without substantially changing benign inputs.

- **Ensemble Method:** Its strength lies in combining diverse models, making adversarial attacks difficult [92].

V. THREAT MODEL

To establish a comprehensive foundation for our survey, this section clarifies the scope of adversarial attacks in NIDS, outlining key threats and their practical implications. We explore the main types of adversarial threats, discuss

the focus on evasion attacks, and provide a structured framework to support the subsequent analysis. This approach bridges the foundational definitions presented in Section IV with the survey's central focus on addressing practical and research-relevant adversarial challenges.

A. COMPARATIVE OVERVIEW OF ADVERSARIAL THREAT MODELS

Adversarial attacks on machine learning-based NIDS can be classified according to their timing and nature. This subsection investigates the comparative aspects of these types of attacks, summarized in Table 3 in terms of their prevalence, real-world feasibility, and evaluation challenges.

Poisoning attacks on NIDS have been moderately studied in the literature, primarily because they require access to training data, which is often restricted or proprietary [93]. However, if the adversary has sufficient access over the training data, they can be fairly simple to implement. While existing studies provide valuable insights [94], [95], [96], their real-world feasibility remains limited. Most adversaries lack the insider access or control over data collection pipelines necessary to manipulate datasets effectively. This attack category is most relevant in settings that rely on shared data collection, such as Federated Learning (FL), which is out of the scope of this survey. Interested readers are referred to existing surveys on FL poisoning [97].

Evasion attacks against NIDS are a significantly more common and widely explored category [93]. These attacks are applicable in both gray-box and black-box settings, where adversaries only have oracle access to the NIDS and possess little to no knowledge of its internal characteristics. In contrast to poisoning attacks, evasion attacks do not require control over protected training data to affect the systems' behavior, potentially making them a practical threat in real-world scenarios. Moreover, their stealthy nature strongly correlates with the attacker's goal of evading detection while effectively executing malicious activities. However, as we discuss in our review, evasion attacks are challenging to implement for NIDS, mainly due to feature constraints (see Section VII).

Finally, backdoor attacks are relatively less prevalent in NIDS research compared to other types. This can likely be attributed to a combination of the difficulty of embedding efficient triggers in structured network data, and the lack of common pre-trained NIDS models, which are popular attack vectors in other areas [98]. Moreover, similar to poisoning, backdoor attacks require insider access or control over training data, making them more relevant in distributed learning settings such as FL.

B. RELEVANCE OF EVASION ATTACKS IN NIDS

This survey focuses on evasion attacks on NIDS due to their prevalence in the literature and the real-world applicability of their threat models. Evasion attacks are carried out during

TABLE 3. Comparative analysis of adversarial attack types in NIDS.

Aspect	Poisoning Attacks	Evasion Attacks	Backdoor Attacks
Prevalence in literature	Medium	High	Limited
Implementation complexity	Low	High (especially black-box)	Medium
Real-World applicability	Limited	High	Limited
Common datasets	CICIDS2017, CSE-CIC-IDS2018	NSL-KDD, CICIDS2017	Rarely evaluated in NIDS contexts

the inference phase, requiring minimal access to internal systems and no influence over training data. This threat model aligns closely with the typical approach of external attackers, making these attacks particularly viable in both gray-box and black-box scenarios. By leveraging system feedback, adversaries can systematically craft adversarial examples without direct access to the model, reinforcing their real-world applicability. Additionally, the structured nature of network data in NIDS, characterized by complex interdependencies between features, presents significant challenges in crafting adversarial examples that remain both effective (i.e., evade detection) and realistic (i.e., remain valid network traffic). Investigating evasion attacks within this framework enables researchers to systematically address these challenges, facilitating the development of robust defenses that are not only effective but also practical for real-world deployments. Finally, evasion attacks are widely prevalent in the literature due to their stealthiness and low cost. Their capacity to evade detection without altering operational pipelines poses a direct threat to the NIDS core functionality.

NOTE. This survey focuses on evasion attacks, addressing the urgent need to evaluate their feasibility. These attacks target a variety of defenses and align with real-world threats, requiring neither system access nor training influence. Our analysis ensures both theoretical insights and practical guidance for strengthening NIDS.

VI. LITERATURE REVIEW OF INTELLIGENT NETWORK INTRUSION DETECTION SYSTEMS

Machine learning, particularly deep learning, shows promise in NIDS. While traditional methods (e.g., Support Vector Machines (SVM), Decision Trees (DTs), and Naive Bayes (NB)) are simpler to interpret, they perform poorly against complex attacks. In contrast, DL-NIDS relies on techniques like Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks to learn intricate temporal and spatial patterns, achieving higher detection rates.

This section reviews ML and DL-NIDS by learning approach (single vs. ensemble), analyzing their advantages, disadvantages and their vulnerability to evasion attacks. Table 5 summarizes these aspects, while Table 4 presents limitations contributing to adversarial vulnerabilities. Moreover, key questions are raised regarding their practical deployment and defense effectiveness.

A. SINGLE LEARNING

Single learning refers to a ML approach where a single, monolithic model is used to perform anomaly detection or classification on unseen data.

1) TRADITIONAL MACHINE LEARNING APPROACHES

Research persists on traditional ML methods for NIDS due to their interpretability compared to DL, aiming to fill the performance gap with DNN. For instance, Chkirbene et al. [99] developed two NIDS models TIDCS and its accelerated variant TIDCS-A, adopting a hybrid feature selection approach. Their approach combines randomization, wrapper, and filter techniques to identify critical features. It selects several feature subsets, trains a Random Forest (RF) classifier on each, scores features based on accuracy, and keeps the highest-scoring ones for the final model. Despite the strong classification results, this strategy risks bias by relying on pretrained accuracy for feature selection. Moreover, prioritizing accuracy may decrease robustness, increasing vulnerability to evasion attacks.

Alahmadi et al. [100] propose a bot identification method using Markov Chains and RF classifiers modeling network behavior as state transitions to extract high-level protocol features for botnet detection. Since evasion would necessitate modifying malware logic, this improves robustness. However, similar system call transition techniques have shown susceptibility to evasion [101]. Further research is also necessary due to potential overfitting and the lack of cross-dataset validation.

For detecting DoS attacks in SDN, Ramamurthy et al. [102] propose an in-controller NIDS, incorporating control packets from unseen network flows. It examines IP addresses, ports, and timing using entropy-based anomaly detection with an SVM model. Despite its smooth integration with SDN, its dependence on easily perturbed features makes it prone to evasion attempts.

TAKEAWAY. For optimal performance, traditional ML-based NIDS depend on dimensionality reduction and complex feature selection. Although this increases the focus on classification, it also increases complexity, introduces bias, and decreases generalization to unseen data. Additionally, since robust selection and reduction procedures are difficult to implement, selected features, even though they are descriptive, frequently lack resilience to adversarial attacks.

2) DEEP LEARNING APPROACHES

To overcome traditional ML limitations, recent NIDS research uses DL to learn complex features. This focus is divided into two categories: fully unsupervised and guided feature representation.

a: FULLY UNSUPERVISED FEATURE REPRESENTATION

This family of works uses end-to-end DNN to autonomously learn feature representations from unlabeled data for classification. Zhou et al. [103] developed DFEL, a DL-based NIDS that pre-trains a DNN on historical NIDS data to extract features from IoT traffic. These features are then used to train another DNN for anomaly detection, leveraging the generalist DNN's data representation for classification. Similarly, Manimurugan et al. [104] use a Deep Belief Network (DBN) to extract features from raw network traffic, combining unsupervised pre-training with supervised fine-tuning for IoT attack detection. In line with [103], [104], Alom et al. [105] proposed a DNN-NIDS using neuromorphic computing, combining unsupervised and supervised training. A fully connected network trained on labeled data classifies the features learned by an autoencoder from unlabeled data. For evaluation, the weights of the trained DNN are quantized and mapped to an IBM TrueNorth neuromorphic SoC.

TAKEAWAY. Fully unsupervised feature representation uses unlabeled data to extract key features for domains with limited labeled data (e.g., IoT), improving classifier performance. However, it risks overfitting to source-domain-specific features, leading to biases. Moreover, since loosely limited features are simpler to exploit, relying only on unsupervised learning makes a system more susceptible to adversarial attacks [106].

b: GUIDED FEATURE REPRESENTATION

This family of works uses a combination of DNN and other techniques to learn descriptive traffic embeddings for intrusion detection. Laghrissi et al. [107] use LSTM with PCA and Mutual Information (MI) for feature selection and dimensionality reduction. While MI chooses the most pertinent features based on statistical relationship with labels, PCA converts correlated features into an uncorrelated set. After that, the model is trained for both multiclass and binary classification.

In [108], a CNN-NIDS was built using adaptive particle swarm optimization (APSO) for hyperparameter tuning. Statistical traffic features (e.g., correlation coefficients, mean, variance) are manually extracted from a custom IoT dataset, feeding them into a 1D-CNN for feature learning and classification. In order to evaluate offspring fitness, APSO uses a validation set to optimize the CNN's hyperparameters.

Moreover, Thakkar et al. [109] employ a fusion-based approach, selecting statistical features from raw traffic data based on a ranking system considering standard deviation,

mean, and median. Iteratively, features are added in order of rating until accuracy stops improving.

Based on augmented NetFlow data, Piskozub et al. [110] present MalPhase, a multi-phase method for anomaly detection and malware classification. It integrates expert-driven feature selection with AE-based representation learning, training the AE to reconstruct malware traffic. MalPhase's multi-tiered architecture detects short activity spikes at lower tiers and sustained malware communication at higher tiers.

Additionally, For malicious traffic identification, Zhong et al. [111] combine supervised DNN-based classification, unsupervised feature learning, and handcrafted feature engineering. They extract SDN flow features using Damped Incremental Statistics [112], refine them with an AE, and produce an anomaly score. An LSTM is then trained by anomalous traffic to identify new traffic as either malicious or normal.

TAKEAWAY. Selecting key features while avoiding biases is simpler by combining DL with expert-guided feature selection. However, most approaches ignore feature representation robustness, making DL-NIDS susceptible to sophisticated evasion strategies. Despite learning complex, non-linear data relations and resisting noise, DL-NIDS are still prone to adversarial threats, and their black-box nature limits decision analysis.

B. ENSEMBLE LEARNING

In contrast to single learning, ensemble learning combines the prediction of multiple models to improve performance and resilience, reducing the error rate if model errors are independent and uncorrelated.

1) COMBINING TRADITIONAL MACHINE LEARNING ALGORITHMS

Several works propose combinations of different traditional ML techniques to address the limitations of any individual approach. For intrusion detection, Tama et al. [113] combine feature selection and ensemble classification. Evolutionary algorithms, (e.g., particle swarm optimization), select optimal feature subsets based on decision tree accuracy. The next step is a two-stage ensemble classifier: A bagged ensemble of weak learners is trained using rotating forest filters' feature subsets, and predictions are finalized by majority vote. Similarly, Zhou et al. [114] suggest a two-stage approach with feature selection followed by ensemble classification. They employ Correlation-Based Feature Selection for initial selection and an evolutionary algorithm to eliminate redundant features. However, unlike Tama et al. [113], they use a decision tree and random forest ensemble for classification. Although weak classifier ensembles improve generalization and robustness to data shifts [115], their reliance on performance-based feature selection may induce biases in training data, hence compromising generalization.

In [116], authors propose an intrusion detection method using an SVM ensemble with feature augmentation and train set splitting. Fuzzy c-means clustering partitions training data into overlapping subsets, which are augmented and classified by an SVM ensemble. For final classification, their outputs are subsequently combined using a different SVM model.

TAKEAWAY. Combining traditional ML models boosts performance and generalization, with heterogeneous ones reducing shared errors and improving resilience. However, they increase computational costs, which restricts their application in resource-limited settings. Overreliance on performance metrics for feature selection may present biases, affecting generalization. Homogeneous ensembles share vulnerabilities, making heterogeneous models a more reliable defense against adversarial attacks [92].

2) COMBINING DEEP AND TRADITIONAL MACHINE LEARNING

To overcome the limitations of single-model DL and ML techniques, ensemble models integrate both to balance their weaknesses. Marir et al. [117] propose a distributed intrusion detection framework merging a Deep Belief Network (DBN) for feature extraction with an ensemble SVM for classification through majority voting. Despite its effectiveness across network settings, its multi-layer networks are resource-intensive, difficult to maintain, and complex to tune without adding training data biases.

Moreover, Imran et al. [118] classify network traffic by combining an SVM with a non-symmetrical deep auto-encoder (NDAE). Their encoder-only design eliminates reconstruction, which lowers computing costs. Similar to other AE-based techniques, it improves correlation detection and feature learning [110], [117]. However, it lacks resilience against adversarial attacks, and its reliance on unsupervised learning risks overfitting to training data features.

In another study [119], authors propose an RNN-based feature fusion strategy with two phases: (i) RNN-based feature learning and dimensionality reduction via Kernel PCA, and (ii) classification using an ensemble of RF, SVM, and logistic regression with voting. While effective, this method is computationally demanding and unsuitable for real-time applications. It is also vulnerable to adversarial attacks due to its reliance on temporal correlations [120].

Recently, a two-level ensemble framework is proposed in [121]. First, class predictions and probabilities are generated, then refined using a different ensemble. It combines various models (DNN, DT, KNN, SVM) and boosting methods, creating a highly heterogeneous architecture. While effective across datasets, its high computational cost limits real-time use. Moreover, the complex interplay between classifiers complicates interpretability and may present overfitting and biases from training data.

TAKEAWAY. Heterogeneous DL-ML ensembles enhance intrusion detection, balancing accuracy and scalability. However, unsupervised learning introduces biases, boost adversarial risks, and reduces explainability. Large ensembles also fail in resource-limited settings [119].

3) COMBINING MULTIPLE DEEP LEARNING ALGORITHMS

To improve detection, studies combine multiple DL models to employ DNNs' feature learning and non-linear correlation detection. Mirsky et al. [112] propose Kitsune, an unsupervised AE-based NIDS using Damped Incremental Statistics and AE reconstruction error. It maps network traffic into autoencoders, with a final anomaly score calculated by another AE. While simple to train and implement, its reliance on temporal features makes it susceptible to evasion attacks, a risk amplified by its fully unsupervised nature.

Furthermore, AttackNet was proposed in [122], a CNN-GRU model for classifying network time-series data. It applies a limited number of gates to decrease overfitting while using CNN and GRU to analyze spatial and temporal features.

Additionally, in [123] intrusion detection within IIoT environments was addressed through a hybrid DL strategy combining LSTM and DNN. An LSTM network processes the time series sensor data, while a DNN separately processes non-time series traffic features. A single dense layer is used to combine the LSTM and DNN outputs and perform classification.

Similarly, Nazir et al. [124] rely on LSTM and CNN for anomaly detection in IoT networks. CNN extracts structural features, a pooling layer minimizes dimensionality, and LSTM captures temporal correlations. For categorization, their outputs are combined using a dense layer. Despite its effectiveness, the model is susceptible to adversary manipulation and evasion due to its dependence on temporal correlations.

Another study proposes [125] a two-step DDoS detection architecture using AE, CNN, and LSTM in SDN settings. CNN and LSTM classify the features that a fully connected AE obtains from SDN switch flow table entries and statistics. Relying on unconstrained statistics increases the system's vulnerability to evasion attacks since adversaries can exploit easily manipulated features, even when unsupervised learning improves feature extraction.

TAKEAWAY. DL ensembles excel in classification and generalization by learning descriptive features from unlabeled data. Heterogeneous models capture spatial and temporal patterns but lack explainability and remain vulnerable to adversarial attacks without robust defenses.

TABLE 4. Strengths, Limitations, and Practical implications of ML and DL-based NIDS.

Approach	Strengths	Limitations	Practical Implications
Traditional ML	Simpler models (e.g., SVM, DT, NB) with lower computational requirements.	Difficulties in addressing high-dimensional data and complex attack patterns.	Suitable for low-resource environments but inadequate for evolving threats.
DL-Based NIDS	High accuracy due to techniques like CNNs and LSTMs for spatial and temporal pattern recognition.	Require GPUs for real-time processing, resource-intensive, and lack interpretability.	Effective for modern attacks but challenging to deploy in edge environments and dynamic networks.
Ensemble Models	Combine multiple algorithms for higher detection accuracy and robustness.	Increased complexity and computational overhead; synchronization challenges between components.	Promising for large-scale networks but require careful balancing of scalability and efficiency.

C. DISCUSSION

Both traditional ML and DL methods have shown promise in NIDS. In situations when features are expertly defined, traditional ML models perform well, providing better explainability and classification confidence. However, their reliance on static features, restricts their ability to adapt to novel threats. In contrast, DL models autonomously learn complex patterns and generalize well to unseen attacks but have significant challenges like high computational requirements, dependence on black-box architectures, and limited adaptability to concept drift. Ensemble models combine the strengths of several approaches to boost robustness but present synchronization issues and processing expense. Table 4 summarizes the strengths, limitations, and practical implications of these approaches. Although they improve detection capabilities, their limitations make them prone to adversarial exploitations. For instance:

- **Traditional ML-based NIDS** rely on static features, that allow attackers to create surrogate models and avoid detection.
- **DL-based NIDS** are more vulnerable to adversarial examples due to their reliance on opaque decision-making and high-dimensional feature spaces.
- **Ensemble models**, despite their robustness, require resource-heavy computations, which attackers can exploit to degrade performance or evade detection.

Addressing these challenges raises critical questions about the vulnerabilities of ML and DL-based NIDS to adversarial attacks:

RAISED QUESTIONS.

- **Question 1.** How can adversarial attacks exploit the structured data constraints inherent in NIDS?
- **Question 2.** Do ensemble techniques provide sufficient resistance to adversarial attacks to justify their computational costs?
- **Question 3.** Do existing approaches balance accuracy, explainability, and resource efficiency to improve real-world applicability?

The following sections critically examine how adversarial attacks exploit these vulnerabilities and evaluate the current state of research on defenses for intelligent NIDS. By analyzing these questions, we aim to identify advancements and unexplored gaps for developing robust NIDS capable of countering sophisticated adversarial strategies.

VII. ADVERSARIAL THREATS TO NETWORK INTRUSION DETECTION SYSTEMS: CRITICAL REVIEW AND PRACTICAL DIRECTIONS

Evasion techniques, such as adversarial examples, present a fundamental threat to intelligent NIDS. However, unlike domains using unstructured data (e.g., image/audio classification, NLP), generating adversarial examples against NIDS is more challenging due to the interdependency and constraints between features in structured data. Network traffic follows specific protocols with predefined features, data types, and relations between different packet fields, making it difficult to manipulate the traffic coherently while also generating effective attacks. Adversarial manipulations must maintain the semantic validity of the data (e.g., changing the protocol field to a non-existent value) and its underlying structure (e.g., changing a packet header format).

This section presents an analytical study of existing research on methods to generate adversarial examples that evade ML-NIDS, classified into white-box, black-box and gray-box attacks (summarized in Table 6). By systematically evaluating these approaches, we analyze their theoretical contributions while critiquing their practicality in real-world settings. This analysis identifies key limitations, such as the challenge of adhering to feature constraints and the reliance on unrealistic threat models, which are further explored in the end of this section (i.e., VII-D)

A. GENERATION OF WHITE-BOX ATTACKS

In white-box attack settings, adversaries have full knowledge of the target NIDS, including architecture and model parameters. While such a level of access might be unlikely in real-world scenarios [93], white-box attacks provide valuable insights into potential attack vectors and vulnerabilities.

TABLE 5. Comprehensive summary of recent papers for the enhancement of machine learning and deep learning based NIDS.

Ref.	Year	Feature Selection	Learning Method	Algorithms	Dataset	Pros	Cons
[99]	2020	Model-based importance score	Single learning	ML (DT & RF)	NSL-KDD & UNSW	Low false positives	Feature selection bias, lack of robust features
[100]	2020	Markov Chain	Single learning	ML (RF)	ISCX, Stratosphere IPS, CTU-13	Privacy-preserving, unseen detection	Botnet-only, weak to obfuscation/reordering evasion
[102]	2019	Manual (entropy)	Single learning	ML (SVM)	Mininet simulation	Transparent, integration with SDN controller	DoS-only, easily manipulated features
[103]	2018	DFEL	Single learning	DL (DFNN)	NSL-KDD & UNSW-NB15	Time-saving applications	Inadequate capacity to reduce data dimensionality
[104]	2020	DBN	Single learning	DL (DBN)	CICIDS 2017	Effective detection rates, potential for scalability	Evasion through feature deletion
[105]	2017	AE	Single learning	DL (DNN)	NSL-KDD, IBM NSCS & IBM TrueNorth	Potential for scalability	Overfitting
[107]	2021	PCA+Mutual Information (MI)	Single learning	DL (LSTM)	KDD'99	Great capability of learning	Overfitting, easily manipulated features
[110]	2021	AE	Single learning	DL (MLP)	ClamAV, Ember, GT Malware, MalRec, MalShare, VirusShare, VX Underground & Miscellaneous	High detection rate, robustness to noise	Sensitivity to advanced adversarial attacks
[108]	2021	Manual	Single learning	DL (CNN)	Public IoT data set created by [126]	Multi-type attacks detection	Biased feature selection
[109]	2023	Sequential Forward Selection (SFS) with statistical ranking	Single learning	DL (DNN)	NSL-KDD, UNSW-NB15, & CIC-IDS-2017	Informative feature selection	Sensitivity to outliers
[117]	2018	DBN	Ensemble learning	DL & ML	KDD'99 CUP, NSL-KDD, UNSW-NB15, & CICIDS2017	Robust to imbalanced data	Sensitivity to outliers
[118]	2022	Non-symmetric deep AE	Ensemble learning	DL & ML (stacked (NDAE)& SVM)	KDD CUP'99	Less susceptible to overfitting	Missing subtle features
[119]	2022	Implicit feature learning	Ensemble learning	DL (RNN, LSTM, & GRU) & ML (RF, SVM, & LR)	KDD-Cup'99, SDN-IoT, UNSW-NB15, WSN-DS, & CICIDS-2017	Automated feature learning, capturing temporal patterns	Overfitting, difficult interpretation
[113]	2019	Hybrid (Evolutionary algorithms+REP Tree)	Ensemble learning	ML (Rotation Forest & bagging)	NSL-KDD & UNSW-NB15	Robust feature selection	Sensitivity to outliers
[114]	2020	Hybrid (CFS+Evolutionary approach)	Ensemble learning	ML (C4.5, RF, & Forest by PA)	NSL-KDD, AWID, & CIC-IDS2017	Robust feature selection	Sensitivity to feature correlations
[116]	2019	Feature augmentation	Ensemble learning	ML (DT-EnSMV)	NSL-KDD	Handle the complex feature interdependencies	Limited feature control
[121]	2024	XAI & Information gain	Ensemble learning	DL & ML (DNN, DT, KNN, MLP, AdaBoost, Logistic Regression, SVM, & Extreme Gradient Boosting)	RoEduNet-SIMARGL2021, NSL-KDD, & CICIDS-2017	Improved individual and ensemble performance	Limited outlier detection
[122]	2024	Implicit feature learning	Ensemble learning	DL (CNN & GRU)	N_BaIoT	Less susceptible to overfitting, simple architecture	Sensitivity to outliers, missing important features
[123]	2022	Implicit feature learning	Ensemble learning	DL (LSTM & DNN)	N_BaIoT	Scalability and generalizability for handling large datasets	Limited feature control, sensitivity to noisy data
[124]	2024	Implicit feature learning	Ensemble learning	DL (CNN & LSTM)	N_BaIoT & CICIDS2017	Scalability and generalizability	Overfitting, sensitivity to outliers
[125]	2022	AE	Ensemble learning	DL (CNN & Bidirectional Long Short-Term Memory)	ISCX-IDS-2012 & UNSW2018	Improved DDoS attack detection	Overfitting, misinterpretation by CNNs and BDLSTMs, sensitivity to outliers

1) VULNERABILITY OF DL-NIDS

The application of traditional gradient-based attacks against DL-NIDS has been analyzed by several studies in the literature. In [127], the authors analyze the vulnerability of a DNN-based NIDS against JSMA [128]. The authors explore all possible combinations of network traffic features and employ a modified saliency map to prioritize the features that minimize the true class probability. Effectively, through the modified saliency map, the attack determines the most sensitive feature combination to craft adversarial perturbations.

Similarly, in [129] Ayub et al. study the vulnerability of MLP-NIDS to JSMA. While they highlight the effectiveness of the attack, these studies do not consider the interdependencies and constraints between traffic features. This limits the applicability of the findings, as the authors do not verify whether the adversarial perturbations respect the semantic validity of the data, or whether they affect the expected behavior of the network traffic in a realistic scenario.

Pawlicki et al. [130] explored several adversarial attacks such as CW [74], FGSM [35], BIM [72], and PGD [131] against DNN-NIDS. All the adversarial attacks were grouped

into a single attack class for the IDS training and testing. The findings highlight the effectiveness of different gradient-based adversarial attacks in causing misclassification. However, since adversarial attacks were applied without modification, the generated perturbations did not respect the semantic validity of the network traffic (e.g., negative packet size). The authors work around this issue by zeroing out invalid fields. However, such an approach still does not guarantee the semantic validity of network packets, potentially resulting in perturbed features that cannot be achieved in a real network setting. Furthermore, the authors do not discuss feature interdependencies and do not consider feature constraints in their attacks.

In [132], the authors study how correlations between network traffic features impact the effectiveness of adversarial attacks that were evaluated against two types of DL-NIDS:

- *Packet-based NIDS*: The proposed attack relies on a local copy of the target NIDS to identify packet-level perturbations that lead to misclassification. The authors define a set of legitimate packet transformations that maintain the semantic validity of the data. Differently from previous works, they do not use gradient-based techniques but rely on splitting, delaying, and injecting packets in the network.
- *Flow-based NIDS*: To consider feature interdependencies and constraints at the flow level, the authors categorize the features into four groups based on how freely each feature can be perturbed while preserving consistency. During the attack, three distinct masks are employed to identify which group of features to modify. The attack proceeds through multiple iterations of perturbations, progressively expanding the permissible range of modifications to generate the adversarial examples.

Searching for an adversarial perturbation with a lower score can create a buffer zone between perturbations and real-world traffic, especially for dependent features outside the attacker's control. However, variations in real-world attack traffic might cause the NIDS score to fall outside this buffer zone, leading to detection. This highlights the challenge of accurately simulating the dynamic properties of real-world network traffic.

2) VULNERABILITY OF TRADITIONAL ML-NIDS

Several attacks have been proposed against Traditional ML-NIDS. Anthi et al. [133] studied the effectiveness of JSMA against ML-NIDS based on RF and J48 classifiers in ICS environments. Since RF and J48 are non-differentiable algorithms and JSMA requires gradient computation, the authors rely on a pre-trained DNN that is used as a proxy to generate the adversarial perturbations instead of the target the ML-NIDS. While relying on a differentiable proxy model to generate JSMA shows promising results, the authors do not address concerns related to the transferability of the attacks to the original models, nor the problem of feature consistency and semantic validity of the perturbed samples.

In another study, Alhajjar et al. [134] investigated the sensitivity of different ML-NIDS (e.g., bagging, gradient boosting, linear discriminant analysis, and quadratic discriminant analysis) to adversarial manipulations. They used Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) to iteratively modify malicious traffic data and employed a Generative Adversarial Network (GAN) to generate adversarial samples. The GAN's generator creates adversarial samples, while the discriminator tries to distinguish them from real traffic. To respect feature constraints, the authors relied on expert knowledge to identify features that may be modified without disrupting network functionality. While the proposed approach shows promising results and respects feature constraints, feature interdependencies and semantic validity were not considered. Moreover, the approach is computationally expensive, limiting its applicability to real-time attacks.

TAKEAWAY. Intelligent NIDS are confirmed to be vulnerable to white-box adversarial attacks, but most studies ignore practical factors. Moreover, the transferability concept [133] remains uncertain because, in real world settings, surrogate models may not accurately mirror the target model's behaviour.

B. GENERATION OF BLACK-BOX ATTACKS

Black-box attack settings assume that the attacker has no knowledge of the details of the target NIDS, such as architecture or parameters. Compared to white-box attacks, black-box settings provide a more realistic threat model and are more representative of real-world scenarios. Typically, black-box attacks rely on an "oracle" model that adversaries can query to validate whether the NIDS detects the perturbed traffic. This feedback from the oracle model is utilized to further refine the perturbation until it achieves evasion. Figure 5 illustrates a general framework for these attacks.

Common techniques include surrogate models, which approximate the decision boundary of the target NIDS, and iterative query-based techniques, which refine perturbations based on feedback from an oracle. Although these techniques shed light on the practicality of black-box attacks, they rely on concepts like unrestricted access to oracles and accurate surrogate models, which are covered in detail in the following subsection.

1) VULNERABILITY OF DL-NIDS

In [135], the authors evaluate C-LSTM, an ensemble model combining CNN and LSTM, and individual DNN and CNN models against a variety of black-box attacks: Natural Evolution Strategies (NES) [136], Boundary Attack [137], HopSkipJumpAttack [138], Pointwise Attack [139], and OPT-Attack [140]. The authors assessed various aspects of adversarial sample generation, including attack success rate, query efficiency, and transferability. The results of the study highlight the effectiveness of these attacks against

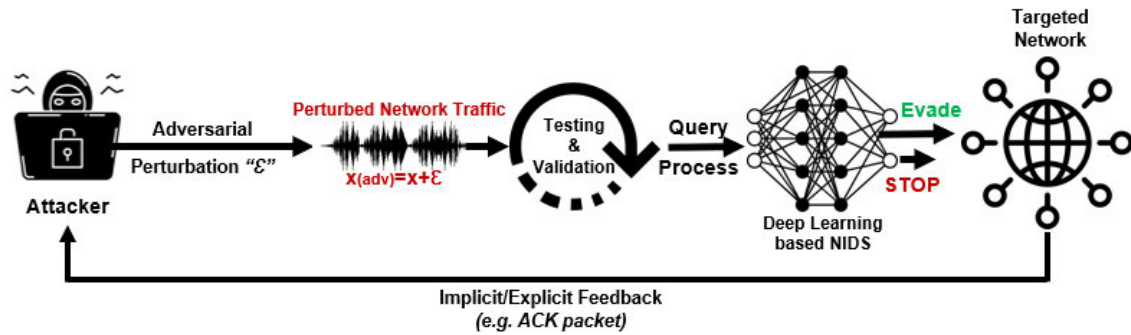


FIGURE 5. Adversarial perturbation against neural network based-NIDS (Black-Box).

both single-model and ensemble NIDS. However, the study does not properly address concerns related to feature interdependencies and constraints. Moreover, crucial details of the attacks related to feature selection and the query process were not covered, making it difficult to evaluate the real-world applicability of the findings.

In addition, the authors in [141] explore the creation of adversarial DoS attacks aimed at evading DNN-NIDS. Their analysis reveals that traditional constraints for generating adversarial examples, which rely on distance metrics such as Lp norms, are ineffective when used for DoS traffic. These metrics treat all features equally, despite DoS traffic containing interconnected features with highly variable value ranges. To address this limitation, the authors utilize Mahalanobis distance, which accounts for feature scale differences, enabling more accurate similarity measures between normal and adversarial DoS traffic. While employing different constraints on perturbations helps to better capture individual feature constraints, the approach still overlooks feature interdependencies and semantic validity of the perturbed traffic.

Wu et al. [39] propose a deep reinforcement learning approach consisting of a black-box botnet NIDS and a flow manipulation module interacting with a Deep Q-Learning agent. For each network flow, the agent selects a modification action from a predetermined set that changes communication patterns without affecting functionality. The modified flow is evaluated by the detection model, and the agent is rewarded based on whether or not it is detected as malicious. This feedback loop enables the agent to iteratively refine the learned perturbation strategy and maximize evasion. While relying on a predefined set of modifications ensures that feature constraints and interdependencies are satisfied, it significantly restricts the perturbation space that the agent can explore, thereby limiting the overall effectiveness of the attack.

In [142], TANTRA is an evasion attack that exploits temporal correlations within network traffic. By using benign inter-packet arrival times to train an LSTM model, it generates malicious traffic that imitates these patterns while maintaining the integrity of the packet content. Although this method can fool NIDS that only use timing features, its

effectiveness is limited since most NIDS use additional flow and packet-level information for detection.

Debicha et al. [143] focus on evading flow-based NIDS targeted at botnets. Unlike previous works, the authors train a surrogate model on benign traffic as a proxy for the real NIDS and generate adversarial perturbations relying on the output of the surrogate model. The authors ensure that feature constraints and semantic validity are satisfied through feature masking and projection operations. Feature masking allows perturbation only of features that can be freely modified (e.g., total number of packets), while the projection ensures that the domain of each feature is respected. The attack relies on the transferability of the adversarial attack between the surrogate model and the target NIDS.

Similarly, Zolbayer et al. [42], developed NIDSGAN, a GAN-based attack that relies on a surrogate model to generate the adversarial perturbation. The GAN is trained using the surrogate model as a discriminator, and limitations on the samples generated by the GAN are imposed to maintain the semantic validity of the traffic. While using a surrogate model simplifies the generation of the adversarial perturbation, the effectiveness of the attack largely depends on how closely the decision boundary of the surrogate model resembles the original one. Moreover, constraining the features that can be modified reduces the overall perturbation space, potentially limiting the effectiveness of the attack by restricting its ability to find optimal adversarial examples.

2) VULNERABILITY OF TRADITIONAL ML-NIDS

Similar to [42], IDSGAN [37] employs a GAN architecture and a surrogate model to generate adversarial traffic. The surrogate model is used as a distinguishing factor for the GAN training, providing a differentiable approximation of the target ML-NIDS. The proposed attack relies on a restricted modification mechanism to ensure that only non-functional features are altered for each type of attack, ensuring feature constraints and semantic validity. While the evaluation of the approach shows promising results, it is based on outdated NSL-KDD dataset, which does not accurately reflect modern real-world network traffic. Furthermore, limiting the allowed perturbations to non-functional features further restricts the attack's effectiveness.

TAKEAWAY. Existing research mentions that black-box attacks pose a significant threat to NIDS, yet their full impact remains unclear, as many studies fail to consider real-world limitations [93]. Most papers only partially address feature constraints and semantic validity in perturbed traffic, reinforcing the need for stronger NIDS defenses against emerging attack strategies.

C. GENERATION OF GRAY-BOX ATTACKS

In gray-box threat models, the attacker has partial knowledge of the features and architecture of the NIDS, allowing adversarial manipulations that are better targeted against the specific NIDS compared to the black-box setting. Assuming the attacker knows the feature set used by the target NIDS, Vitorino et al. [144] propose the Adaptive Perturbation Pattern Method (A2PM). A2PM generates adversarial examples by analyzing features to identify valid ranges and correlations. It applies interval patterns to perturb numerical features within limits and combination patterns to preserve coherence in categorical features. These patterns are sequenced to introduce perturbations that comply with domain rules and class-specific requirements.

Sheastsley et al. [145] propose two approaches to generate adversarial attacks that can adapt to specific network constraints:

- *Adaptive JSMA (ASJMA)*: ASJMA incorporates network traffic constraints into JSMA. The attack relies on the concept of primary features, which are features that, when set, limit the permissible range of other features. When generating perturbations, ASJMA verifies whether a given feature is affected by another primary feature and limits the allowed perturbation according to their specific interdependencies.
- *Histogram Sketch Generation (HSG)*: HSG learns meta-perturbations, called *sketches*, from a set of previously generated perturbations. HSG utilizes the results of previous attacks to detect frequently perturbed features and their modifications to learn a generalized attack pattern. The learned sketch can then be applied to new network data to evade detection.

While these techniques result in perturbations that satisfy feature constraints and interdependencies, they require extensive knowledge of features and target network traffic distribution, which may be limiting in real-life settings.

In [146], the authors focus only on evaluating the robustness of ML-based ensembles against various well-known adversarial attacks. To enable the computation of gradient-based attacks on non-differentiable ML models, the authors employ a surrogate DNN model. The adversarial perturbation is computed using the surrogate model, with the success of the attack depending on the transferability of the generated perturbations to the target NIDS. While this paper approved the vulnerability of shallow models to adversarial samples, the original accuracy of the latter has

not significantly dropped compared to other existing attacks (e.g., [134]). Additionally, further information about how the methodology considers the attacker's knowledge of the feature constraints or the data collected during the attack is not provided. Hence, this study does not reflect real-world conditions.

TAKEAWAY. Gray-box attacks balance the complexity of black-box attacks and the limited employment of white-box attacks, presenting a threat to intelligent NIDS. However, in practice, their effectiveness and the balance between attacker knowledge and realistic perturbations remain unclear, requiring further research to evaluate risks and develop robust defenses.

D. FROM THEORY TO PRACTICE

This review identifies significant gaps in the practical applicability of adversarial attacks within network traffic, particularly due to the structured and constrained nature of network data. Unlike unstructured domains (e.g., image or text classification), network traffic features are interdependent and governed by protocol rules. Successful adversarial attacks do not only evade detection, but also maintain semantic validity to avoid generating unrealistic or invalid traffic.

In the following, we explore three main challenges that the practical adoption of adversarial attacks.

1) Feature Constraints and Interdependencies:

Adversarial attacks against NIDS have to consider different feature constraints and interdependencies to generate valid and realistic adversarial examples. These constraints include preserving protocol consistency (e.g., ensuring that protocol types remain within valid categories) and maintaining semantic connections between features (e.g., ensuring that changes to packet size align with the corresponding duration). However, Many existing adversarial strategies fail to fulfil these requirements, resulting in invalid traffic that violates protocol rules or disrupts feature dependencies. For example, changing “packet size” without considering its relationship to “duration” can lead to incorrect traffic patterns. Similarly, failing to consider protocol-specific constraints can result in malformed headers or invalid fields. Practical techniques, such as packet splitting or delaying [147], seek to deal with these issues but are heavily dependent on specific network environments and traffic conditions. To address these challenges, the following methods have been identified as critical:

- **Protocol Validity:** Incorporating domain-specific pre-processing steps or constraints ensures that perturbed traffic considers protocol rules. For instance, invalid fields or malformed headers can be avoided using packet validation techniques [148].

- **Feature Interdependencies:** Methods like adaptive perturbation frameworks and feature-aware regularization ensure alterations to a specific feature (e.g., “packet size”) stay consistent with others (e.g., “duration”), maintaining realistic traffic patterns [149], [150].
- **Semantic Validity:** Optimization-based methods that integrate domain expertise into the attack generation process ensure that adversarial examples remain realistic and efficient, which corresponds to the expected network behaviour [132].
- **Real-Time Applicability:** To ensure applicability for real-world NIDS settings, lightweight adversarial frameworks and hybrid adaptive perturbation techniques achieve a balance between reducing computational overhead and maintaining feature constraints [151].

TAKEAWAY. To overcome protocol, feature, and semantic restrictions, domain-specific adaptations must be prioritized in the development of adversarial strategies for NIDS. Moreover, lightweight techniques should be investigated in future studies to ensure real-time applicability in dynamic network scenarios.

2) Unrealistic Assumptions in Threat Models:

When operating within black-box settings, adversaries rely on querying an oracle function $g(\mathbf{x})$ to evaluate detection probabilities. The optimization goal can be formulated as:

$$\text{Minimize: } \mathcal{L}(\mathbf{x}^*, g(\mathbf{x}^*)), \quad \text{subject to: } \|\delta\|_p < \epsilon,$$

where $\mathcal{L}$ is a loss function representing the detection score, $\|\delta\|_p$ is the perturbation magnitude, and ϵ is the authorized threshold.

Most black-box approaches assume unlimited oracle access to $g(\mathbf{x})$, allowing iterative queries to refine δ . However, in practical settings, query limitations and the detection of probing activities severely constrain this process. For instance, surrogate models trained to approximate $g(\mathbf{x})$ often fail to depict the true decision boundary. Therefore, it reduces the effectiveness of transferability-based attacks.

RECOMMENDATION. Development of attack strategies that minimize oracle dependence, such as one-shot attacks [152] or heuristic approaches [153], and improve surrogate model training through the inclusion of domain-specific constraints.

3) Practical Evaluation of Adversarial Examples:

Adversarial examples have to consider domain-specific constraints to remain realistic and practical for evaluation. This involves ensuring that features such as protocol types and packet sizes adhere to semantic

consistency and fall within valid ranges. However, many studies rely on outdated datasets (e.g., NSL-KDD) that do not represent modern traffic patterns, leading to evaluations that lack generalizability [37], [134]. Furthermore, it is often observed that generated adversarial examples fail to maintain the intended malicious behavior or align with real-world traffic complexities, limiting their practical applicability.

RECOMMENDATION. Adoption of contemporary datasets that exhibit a diverse traffic patterns and attack scenarios. Concurrently, incorporate domain-specific constraints within the evaluation pipeline to ensure the production of realistic adversarial examples.

In summary, addressing the practical challenges of adversarial attacks on NIDS necessitates realistic threat models, robust evaluation methodologies, and respecting feature constraints inherent in network traffic. Focusing on these aspects will allow future research to translate theory into practice, ensuring solutions are both operationally feasible and resilient to dynamic network conditions.

VIII. ADVERSARIAL DEFENSES TO NETWORK INTRUSION DETECTION SYSTEMS: CRITICAL REVIEW AND PRACTICAL DIRECTIONS

Although the feasibility of adversarial attacks on NIDS is limited in real-world scenarios, defensive techniques remain mandatory for fixing known vulnerabilities and preparing for new threats. This section reviews existing defenses in the literature, categorized into adversarial training and architecture reinforcing (As previously presented in Section IV). Table 7 summarizes their strengths and limitations, providing insights into their real-world applicability.

A. ADVERSARIAL TRAINING

Adversarial training is a widely studied defense strategy for enhancing the resilience of intelligent models [85]. By being exposed to adversarial examples during training phase, the model learns to recognize and resist such perturbations.

Alahmed et al. [154] propose an adversarial defense using RF classifiers for NIDS. Their approach includes training a Generative Adversarial Network (GAN) to generate realistic adversarial network traffic examples. The NIDS model is then trained in two phases: first on real traffic data, and subsequently on a combined dataset that includes both the adversarial examples produced by the GAN and the original data. This two-step training process bolsters the model's defenses against adversarial perturbations. The approach is further enhanced through Principal Component Analysis (PCA) for feature selection, and the model's resistance is evaluated using a black-box attack technique. Despite its strengths, this approach has various limitations. Firstly, the reliance on GANs to produce synthetic network data is costly

TABLE 6. Summary of adversarial attacks on NIDS.

References	Year	Type of Attack			Applicability		Limitation
		White-Box	Black-box	Gray-Box	Target Model	Dataset	
[127]	2023	✓	✗	✗	MLP	CIC-IDS2017, CIC-IDS2018, CIC-DDoS2019	No consideration of feature interdependency, lack of realism
[129]	2020	✓	✗	✗	MLP	CICIDS 2017, TRAbID 2017	No consideration of feature interdependency, lack of realism
[106]	2020	✓	✗	✗	DNN	CICIDS 2017	No consideration of feature interdependency Introduction of negative values into certain network traffic features
[132]	2019	✓	✗	✗	DNN	CICIDS 2017	Generalizability and lack of realism
[133]	2021	✓	✗	✗	RF, J48	Power System	Lack of details on handling feature correlations Limited generalizability due to pre-trained model Difficulty in balancing feature modification
[134]	2021	✓	✗	✗	SVM, RF, DT, NB, MLP, KNN, LR, GB, Bagging, LDA, QDA	NSL-KDD, UNSW-NB15	Impact on real-time attacks
[135]	2022	✗	✓	✗	C-LSTM, MLP, CNN	CSE-CICIDS 2018	No explanation on feature selection and correlations Limited information on the attacker's query process
[141]	2019	✗	✓	✗	DNN	KDDcup99, CICIDS2017	Limited effectiveness of traditional distance metrics Lack of details on the optimal settings for Mahalanobis distance parameters Lacks of details on how the attacker gathers information about the target model
[39]	2019	✗	✓	✗	CNN & DT	CTU13	Limited action space exploration Inability to directly exploit feature correlations
[143]	2023	✗	✓	✗	DNN	Ad-Hoc	Subtle, detectable perturbations Lack of explanation on how the attacker gathers information
[42]	2022	✓	✓	✗	DNN	NSL-KDD, CICIDS 2017	Using limited data for the surrogate model, reduce the attack success rates
[37]	2022	✗	✓	✗	SVM, MLP, KNN, RF, LR	NSL-KDD	Outdated data, lack of realism
[142]	2022	✗	✓	✗	LSTM	CIC-IDS2017, Kitsune	Complex real-time setup (less flexible)
[144]	2022	✗	✗	✓	MLP, RF	CIC-IDS2017, IoT-23	If the training data used by A2PM is inadequate, the generated attacks will be unrealistic and detected
[146]	2022	✓	✓	✓	AdaBoost, Bagging, Gradient Boosting, LR, DT, RF, SVC	NSL-KDD, UNSW-NB 15	No consideration of feature interdependency, lack of realism
[145]	2022	✓	✗	✓	ML (Not Specified)	NSL-KDD, UNSW-NB 15	Outdated data, lack of realism

and inefficient. GANs are shown to be tricky to train and stabilize. Secondly, the GAN-generated network data are not validated as real network traffic flow data. The fact that GANs can generate data from a given distribution does not mean that the generated data represents real-world traffic flows.

Similarly, Debicha et al [43] aim at strengthening a DL-based NIDS through the use of adversarial training. To generate the adversarial examples, they rely on three off-the-shelf adversarial generation techniques, namely the FGSM, PGD, and BIM. Similarly to the previous approach, the proposed technique does not consider the constraints required for the generated adversarial network flow to be realistic. Additionally, it shares the same limitation of having limited practical application.

Venturi et al. [155] address network intrusion detection by monitoring network traffic through a graph representation, including both topology and individual connections. The approach uses a Graph Neural Network (GNN) within an AE to condense graph complexities into embeddings that capture important components of each flow and their interactions, acting as a feature extractor. To improve resilience against data perturbations, they rely on adversarial training as well, exposing the AE to both legitimate and malicious traffic to learn robust embeddings. These embeddings, containing network architecture and flow details, are then processed by an RF classifier for normal and abnormal traffic classification. A key limitation to this procedure is its high computational cost, especially in resource-constrained

environments; therefore, optimizing this process for greater efficiency represents a viable direction for building adversarially robust NIDS.

Another approach, the EIFDAA framework [156], provides active defense through adversarial training based on three steps. First, a simulation of attacks in which adversarial examples are generated using BIM, PGD, FGSM, and DeepFool. These attacks modify legitimate data to create malicious samples that mimic typical traffic patterns and evade IDS detection. The second step is adversarial evaluation, which requires including original attack data that represents real attacks together with simulated attacks into the IDS being evaluated. A significant decrease in attack detection indicates an IDS vulnerability. Finally, adversarial training allows the IDS to learn attacker techniques and fortify its defenses against future attempts to exploit similar vulnerabilities by retraining it using a combination of adversarial examples and the original training data. EIFDAA offers a comprehensive evaluation and defensive method. However, its performance depends on the accuracy of the evaluation data and the quality of attack simulations.

Recently, Debicha et al. [143], aimed at crafting a filter-like system to prevent adversarial examples from reaching and fooling the NIDS. This detector acts as a pre-filter using three techniques: adversarial training, adversarial detection, and ensemble methods. Three MLPs are trained in parallel; each one focuses on a specific set of network traffic features based on the possibility

of manipulating them by the attacker (modifiable or unmodifiable). The training data is composed of two classes: adversarial and clean. During detection, each MLP monitors an incoming data point. Each prediction is then weighted based on its training accuracy. The weighted predictions from all three MLPs are integrated using a combination of contextual discounting and Bayesian fusion, leading to a final classification of each data point as either *adversarial* or *clean*. This classification determines whether the data point is permitted to pass through to the NIDS. While effective, running multiple models can be computationally expensive, and accurately classifying features may be difficult due to manipulation by attackers.

TAKEAWAY. Adversarial training must address overfitting and effective generation strategies. Excessive perturbations can deceive decision boundaries and reduce generalisation. In NIDS, resilience depends on generating realistic traffic samples within feature constraints.

B. ARCHITECTURE REINFORCING

The goal of architecture-reinforcing defenses is the creation of ML-based models that are inherently robust to adversarial attacks. These methods often include ensemble models, hierarchical structures, or robust feature processing.

Jiang et al. [157] rely on multi-model and task modelling to achieve adversarial robustness. They analyze network data and divide features into three groups: time-related, packet-related, and state-related, which are less likely to be altered by attackers. Each group is processed by a separate DL model, and the predictions are combined to decide if a flow is malicious. This approach shows good adversarial robustness, while maintaining adequate detection rates in the absence of adversarial attacks. However, its effectiveness depends on accurately predicting which features attackers will target, which can reduce its reliability when adversaries alter their attack.

Jmila et al. [146] assessed the robustness of Gaussian data augmentation against adversarial attacks by introducing random Gaussian noise into the training data. This method aims to expand the dataset and limit the ability of attackers to craft targeted adversarial examples, making the IDS less vulnerable to feature manipulation. The authors compared IDS classifiers trained on original versus augmented data, evaluating them on both normal and adversarial test sets. They measured the differences in accuracy and False Negative Rate to assess the model's performance. While this approach enhances IDS resilience, its effectiveness may be constrained by the inability to fully simulate real-world adversarial manipulations.

Zhang et al. [158] propose another method that incorporates three strategies to defend against different types of black-box adversarial attacks, namely ensemble voting, ensemble adversarial training and adversarial sample

detection, with each strategy having its limitations. For instance, combining multiple deep learning models can increase complexity and heighten the risk of misclassification. Adversarial training, while useful, relies heavily on the attacker's knowledge and risks overfitting. Query detection requires careful threshold tuning and remains vulnerable to attacks that can bypass the initial detection phase. Additionally, these methods raise computational costs and may struggle to keep pace with rapidly evolving threats.

In contrast to traditional models that explicitly classify attack types, a two-layer hierarchical defense system was proposed by McCarthy et al. [159]. It uses a coarse layer to group similar attacks (e.g., various DoS attacks) and a fine layer to classify specific types (e.g., Slowloris). This makes it challenging for attackers to inject perturbations that trick both layers. Despite the high computational cost of this approach and the need to define the attack group hierarchy, it results in a robust model that prevents adversarial examples and maintains high classification accuracy in regular traffic.

Recently, Mutation-based detection has proven effective for adversary defense [62] as have approaches that rely on observing adversarial manipulations occurring near the decision boundary [90]. Wang et al. [160] identify adversarial perturbations by detecting inconsistencies between the NIDS's classification and natural data patterns (manifolds). These manifolds represent the original data structure and decision boundary, making it hard to detect perturbations as they share characteristics with both malicious and benign traffic. The effectiveness of this approach relies on the accuracy of the IDS model, leaving it vulnerable to exploitation if the model has any weaknesses. Moreover, this dual approach, involving manifold analysis and decision boundary detection, is more computationally expensive compared to previous detection techniques.

TAKEAWAY. Reinforcing NIDS with generalized defenses boosts adaptability and reduces retraining needs. While enhancing resilience and maintenance, adjustments for specific tasks limit cross-domain applicability. Balancing security, accuracy, and complexity is significant for efficient, task-specific defenses.

C. FROM THEORY TO PRACTICE

Adversarial defenses are crucial to mitigate the increasing threats caused by adversarial attacks. However, practical limitations including computational feasibility, semantic validity, and adaptation to evolving traffic patterns are frequently ignored by current methods. This section provides research-based recommendations for directing future developments in adversarial defenses.

- 1) **Adversarial Training:** Adversarial training, as defined in Section IV in Equation (6), optimizes model parameters Δ to reduce susceptibility to adversarial perturbations by using adversarial examples during the training process. Although this strategy has a strong theoretical

foundation, its practical implementation faces several challenges:

- **Semantic validity of adversarial examples**

Most existing adversarial training methods ignore structured data constraints. For example, manipulating features like “*protocol_type*” or “*service*” need to remain semantically valid within the network protocol. Future strategies must to specifically incorporate these limitations into Δ using domain-specific rules or expert knowledge.

RECOMMENDATION. Defining Δ as:

$$\Delta = \left\{ \delta : \begin{array}{l} \delta \text{ satisfies feature dependencies} \\ \text{and protocol constraints} \end{array} \right\}.$$

Ensuring adversarial examples remain realistic.

- **Computational cost**

GAN-based adversarial example generation, for instance, are computationally expensive (e.g., [154]). Pre-trained adversarial generators or gradient-free techniques are examples of lightweight perturbation techniques that potentially lower costs without affecting robustness.

RECOMMENDATION. Application of heuristic optimization or evolutionary algorithms to approximate:

$$\delta^* = \arg \max_{\delta \in \Delta} \mathcal{L}(f_{\theta}(\mathbf{x} + \delta), y),$$

reducing the complexity of adversarial training.

- **Adaptability to dynamic traffic patterns**

The static datasets used in current adversarial training techniques are unable to mimic the variability of real network traffic. This constraint can be addressed by integrating incremental updates and online learning into the adversarial training pipeline.

RECOMMENDATION. Development of frameworks that periodically update the adversarial training set based on observed traffic patterns:

$$\mathcal{D}_{\text{new}} = \mathcal{D}_{\text{old}} \cup \mathcal{D}_{\text{dynamic}},$$

where $\mathcal{D}_{\text{dynamic}}$ represents unseen samples reflecting emerging traffic.

- 2) **Architecture Reinforcing:** To boost resistance against adversarial perturbations, ensemble approaches and hierarchical systems are frequently employed. However, they have significant challenges in scalability and adaptability:

- **Scalability in ensemble methods**

Several models are combined in ensemble systems to increase robustness:

$$f_{\text{ensemble}}(\mathbf{x}) = \arg \max_k \sum_{i=1}^n w_i \cdot \mathbb{I}(f_i(\mathbf{x}) = k),$$

where w_i is the weight of each model f_i . However, due to the increased processing complexity, real-time deployment is not feasible.

RECOMMENDATION. Application of lightweight, modular ensembles whose components activate dynamically in response to traffic conditions. For example:

$$w_i = \begin{cases} 1, & \text{traffic matches model } i \text{'s feature set,} \\ 0, & \text{otherwise.} \end{cases}$$

- **Adaptability to new threats**

Static architectures are ineffective against sophisticated attack techniques. Hierarchical defenses, which group and refine attack detection, can enhance adaptability:

$$P_{\text{attack}}(\mathbf{x}) = P_{\text{coarse}}(\mathbf{x}) \cdot P_{\text{fine}}(\mathbf{x}),$$

where P_{coarse} groups attacks (e.g., DoS) and P_{fine} detects specific types (e.g., Slowloris).

RECOMMENDATION. Integration of online learning into hierarchical systems to adapt groupings and refine detection thresholds dynamically.

- **Explainability and trustworthiness**

Many architecture-reinforcing methods act as black boxes, limiting adoption. Deployment in operational environments requires explainable models that produce interpretable outputs.

RECOMMENDATION. Implementation of feature relevance score or saliency-based approaches to explain:

$$S_i = \frac{\partial \mathcal{L}(f_{\theta}(\mathbf{x}), y)}{\partial x_i},$$

where S_i is the sensitivity of feature x_i .

Future studies may fill the gap between theoretical developments and the real-world implementation of adversarial defenses by tackling these issues and implementing the recommendations into practice. The transition from theory to practice will ensure that NIDS are reliable, adaptable, and efficient in practical settings.

IX. ADVERSARIAL EXAMPLE PLACEMENT: FOUNDATIONS FOR ROBUST DEFENSES

Understanding the placement of adversarial examples is important for addressing the gap between theoretical attacks and practical defenses. To the best of our knowledge, none of the existing surveys provide researchers with

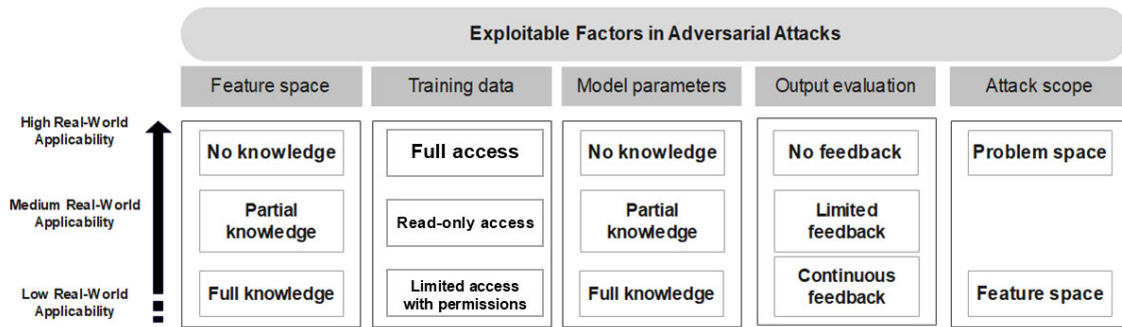


FIGURE 6. Real-world applicability of adversarial factors when generating adversarial attacks.

TABLE 7. Comprehensive summary: Analytical study of adversarial defense strategies to NIDS.

Reference	Year	Defense Method	Strengths	Limitation
[154]	2022	Adversarial Training	High detection rate of FGSM perturbations, while maintaining the original detection level	High training cost and limited generalizability
[43]	2021		Detection for some specific adversarial attacks	Reliance on outdated data and minimal adversarial training
[155]	2023		Strong against specific adversarial attacks	Computational cost, limited generalizability, and limited adversarial training
[156]	2023		Strong against some specific adversarial attack	Limited generalizability and reliance on accurate data
[143]	2023		High detection performance	Computationally expensive, vulnerable to evolving attacks
[157]	2022	Architecture Reinforcing	Improved attack detection through multi-modal analysis	Limited robustness to adaptive and evolving attacks
[146]	2022		Simplicity and improved generalizability	Loss of information, restricted to specific attacks
[158]	2020		High detection performance	Computationally expensive and limited adaptability
[159]	2023		Strong robustness to evasion	Computationally expensive
[160]	2022		High true positive rate with low false positives	Its effectiveness depends on the accuracy of the target model Not adaptive to more sophisticated attacks

insights into the spatial placement of adversarial examples within structured data domains like network traffic, as their main focus on attack generation methods with their defenses. This oversight restricts the understanding of how feature interdependencies and domain constraints shape the space of allowable adversarial examples. In this section, we aim to fill this void, providing a comprehensive analysis and highlighting its implications for NIDS robustness.

A. CATEGORIES OF ADVERSARIAL EXAMPLE PLACEMENT

Adversarial perturbations can be classified based on their relationship to the data manifold [90], as shown in Figure 7.

Let $\mathcal{M}$ represent the manifold of valid network traffic, where $\mathbf{x} \in \mathcal{M}$ satisfies all feature dependencies and semantic constraints.

1) Leave-the-Manifold Attacks

These attacks create perturbations δ such that the adversarial example $\mathbf{x}^* = \mathbf{x} + \delta$ lies outside $\mathcal{M}$:

$$\mathbf{x}^* \notin \mathcal{M}.$$

Researchers have examined this concept, noting that these modifications are more intense and generate features not present in the training dataset [139], [161], [162], [163], [164]. These manipulations are larger and more visible, exploiting the model's weaknesses in handling unfamiliar or out-of-distribution data. For instance, modifying *protocol_type* to an invalid value generates traffic that violates protocol rules, making

these attacks less practical despite their theoretical efficacy.

2) On-Manifold Attacks

On-manifold attacks ensure that the manipulated examples remain within the manifold:

$$\mathbf{x}^* = \mathbf{x} + \delta, \quad \mathbf{x}^* \in \mathcal{M}.$$

This concept states that training data and adversarial samples belong to the same distribution or manifold [90], as illustrated in Figure 7. In other words, these attacks stay within the feature space that the model is familiar with from training. Compared to leave-the-manifold attacks, these perturbations are usually subtle and do not visibly alter the input data. For instance, adjusting timing features (e.g., *duration*) while maintaining consistency with *packet_size* ensures that the adversarial traffic remains valid and undetected by rule-based systems.

3) Low-Dimensional Subspace Attacks

In some situations, adversarial examples are located in a lower-dimensional subspace $\mathcal{S} \subseteq \mathcal{M}$:

$$\mathbf{x}^* \in \mathcal{S}, \quad \dim(\mathcal{S}) < \dim(\mathcal{M}).$$

By exploiting this structure, attackers can find perturbations that significantly lead to misclassifications with minimal input modifications [63]. For instance, targeting only a subset of features like *src_bytes* and *dst_bytes* can effectively evade detection mechanisms.

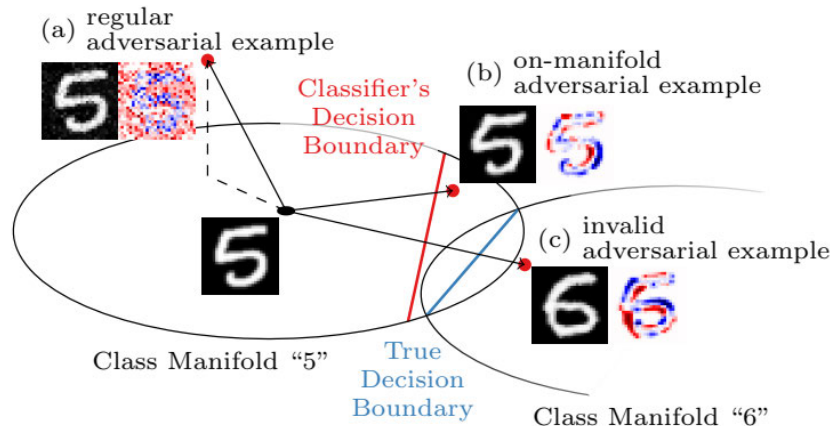


FIGURE 7. Comparative illustration of standard, on-manifold (*Constrained*) Adversarial examples vs. Unconstrained ones [90].

B. IMPLICATIONS FOR NIDS

Studying the placement of adversarial examples within the data manifold has significant implications for the design and resilience of NIDS. This perspective provides different actionable insights:

- **Structured Constraints:** Network traffic data is generally structured, with features that follow specific protocols rules and reflect complex interdependencies. Large perturbations that violate these restrictions are more likely to be detected, as they result in unusual patterns that differ from those of normal traffic. Therefore, in order to avoid detection, adversaries frequently create slight, in-manifold perturbations. By identifying this behavior, NIDS may focus on detecting even minor deviations within the manifold, increasing their susceptibility to sophisticated attacks.
- **Localization for Defenses:** By understanding that adversarial examples tend to lie within the data manifold, researchers can develop strategies that project inputs onto the manifold of legitimate traffic. Methods like manifold regularization can be used to ensure that any deviation from this manifold is examined, thereby identifying potential adversarial manipulations. This approach effectively minimizes the attack surface available to attackers, as off-manifold perturbations can be flagged as suspicious.
- **Adaptive Defenses:** It is easier to create adaptive defense systems that can dynamically react to deviations that are recognized when adversarial placement is understood. For example, to increase future resistance, the system can modify its detection thresholds or retrain on the new data if an adversarial example is found inside the manifold. Given the ongoing evolution of network security threats, this adaptability is crucial.

- **Advanced Feature Engineering:** Feature engineering procedures can benefit from insight into the location of adversarial examples. NIDS designers can prioritize the robustness of features, include extra validation checks, or create composite features that are more resilient to adversarial attacks by determining which features are most vulnerable to these type of manipulations.
- **Real-World Alignment:** In real-world situations, attackers are constrained by the requirement to preserve network traffic's functional validity. It is difficult to cause disruptions that interfere with service or break protocol requirements. By characterizing how adversarial examples behave with the manifold, researchers can develop NIDS that anticipate realistic attack techniques, ensuring that countermeasures are both practical and effective in real-world settings.

By integrating these considerations into the design and implementation of NIDS, their resilience to adversarial attacks can be strengthened, ensuring reliable and secure network protection systems.

X. LESSONS LEARNED AND GUIDELINES

This section explores lessons learned and potential future research directions, providing insights and guidelines to improve future research in the area.

A. THE VALIDITY AND REALISM OF ADVERSARIAL ATTACKS

Lacks of validity and realism limit the deployment of adversarial attacks against NIDS, which reduces its practicality. The key limitations of current research are the following.

1) Unrealistic Threat Models

Adversarial attacks are often evaluated under idealized assumptions about the attacker's capabilities. These include:

- **Perfect Knowledge (White-Box):** Full access to the model's internal parameters and architecture.

TAKEAWAY. From the perspective of security evaluation, white-box assumptions are somehow important even when they fail to accurately reflect deployment facts. These configurations enable researchers to investigate the worst-case scenarios and uncover weaknesses that more constrained models could ignore. Furthermore, since white-box attacks reveal the upper limits of potential adversarial impact and guide the development of more robust detection systems, they frequently serve as the foundation for developing robust countermeasures [165], [166].

- **Unlimited Query Access (Black-Box):** The assumption that an attacker can query the target model without limitations.

However, in real-world implementations, query access is typically limited by rate-limiting algorithms that restrict both the volume and frequency of queries. These limitations hinder rapid, high-volume probing, forcing adversaries to trade off between query efficiency and stealth, as exceeding thresholds may trigger anomaly detection or access denial. While these limitations constrain centralized querying, attackers may attempt to circumvent them by using distributed infrastructures, like botnets, to propagate queries and avoid rate limits. However, due to traffic irregularities and coordination patterns, this strategy adds operational complexity and raises the possibility of detection. Therefore, even when technically possible, large-scale querying remains constrained by the practical requirements of stealth and evasion.

The practical effectiveness of adversarial attacks rely on the attacker's access to and knowledge of several critical factors [93] (as shown in Figure 6):

- **NIDS Feature Knowledge:** Due to system constraints, real-world adversaries lack full knowledge of the features a NIDS considers for classification. Feature selection is usually proprietary, making adversarial feature selection a complex task.
- **Training Data Access:** While some attacks (e.g., model extraction) exploit access to training data, most adversarial examples do not necessitate direct access [143]. However, attackers rely on surrogate models trained on proxy datasets or query-based learning. Research should consider how realistic these assumptions are when evaluating attack feasibility.
- **Model Architecture and Configuration:** The effectiveness of adversarial attacks can be significantly affected by an understanding of the internal structure of the target model. Attackers may have access to model parameters (e.g., weights and layer structures)

in white-box settings. Moreover, hyperparameters (e.g., learning rates, activation functions, and regularization techniques) also influence model behaviour and robustness. Adversarial attack design becomes more complex in real-world situations since complete understanding of these components is rarely available.

- **Output Evaluation and Oracle Access:** Several adversarial attack models assume that attackers can access detailed classifier outputs like confidence evaluations or full probability distributions. However, in operational NIDS deployments, responses are often obfuscated, and access to raw outputs is limited. This significantly limits iterative adversarial optimization in practical applications. Therefore, assuming unrestricted oracle access to NIDS is usually an excessively strong assumption that is incompatible with real-world implementations.
- **Attack Scope and Constraints:** Both the attack's scope and the constraints imposed by effective countermeasures must be considered to accurately assess the feasibility of adversarial attacks in realistic NIDS deployments. There are two main factors that affect how effective an attack is:
 - **Problem-Space vs. Feature-Space Attacks:** Some research simply manipulate feature-space representations without ensuring that adversarial examples remain valid when converted into real network traffic. However, attacks have to maintain semantic validity to be applicable in practical situations.
 - **Rate-Limiting and Detection Evasion:** Many existing studies ignore practical security measures like rate-limiting, anomaly detection, and adaptive defenses. Ignoring these limitations can result in an overestimation of attack success rates and restrict real-world applicability.

TAKEAWAY. Research should consider realistic threat models, query limits, and network feature constraints. Future work must focus on adaptive, stealthy attacks that align with real-world deployment conditions.

2) Preservation of Feature Dependencies

As discussed in Section VII, existing studies often fail to consider the potential impact of feature constraints and correlations when designing adversarial attacks against NIDS. This leads to a gap between theoretical analysis, recommendations, and their practical feasibility. Several factors must be respected:

- **Syntactic and Semantic Validity:** The attack might be invalid if packet headers or content are modified without considering the context of the network communication. Network protocols have well-defined

syntactic and semantic rules that dictate the structure and behavior of traffic. Arbitrary modifications to packet data may lead to violations of these protocols.

TAKEAWAY. Semantic integrity is crucial in adversarial environments, even though many operational NIDS do not yet enforce strong semantic validation because of resource limitations. The absence of such validation, on the other hand, leaves a crucial blind area that attackers can exploit to evade detection. Thus, maintaining syntactic and semantic coherence presents a valuable opportunity to improve NIDS robustness in addition to being necessary for producing realistic adversarial samples. To bridge this gap, we encourage future research to think about incorporating lightweight semantic consistency tests.

- **Time-based dependencies:** Network traffic often presents time-sensitive patterns. If adversarial perturbations incorrectly alter these temporal dependencies (e.g., sending an acknowledgment before receiving the associated data packet), it will lead to protocol violations and lead to failure in real-world settings.
- **Feature Interdependencies:** Modifications to certain attributes, such as packet size or frequency, lead to changes in other features, such as total number of bytes or communication duration. Adversarial attacks and defenses must consider these dependencies to be effective in real-world settings.

3) Limited Exploration of Adversarial Example Placement

As discussed in Section IX, the literature has predominantly focused on the generation and defenses of adversarial attacks, with insufficient attention to their placement within the data manifold. Understanding whether adversarial examples lie on-manifold or off-manifold is essential, as it impacts the effectiveness of detection and defense strategies. This limitation introduces an opportunity for future research to investigate the spatial characteristics of adversarial examples in structured data, leading to more robust NIDS.

Despite the theoretical advances in adversarial attack research, many existing studies are limited by oversimplified attack methods and outdated datasets (e.g., NSL-KDD) that fail to mimic the complexities of modern network traffic. These oversights result in evaluations that lack operational relevance, reducing the practical utility of such findings in realistic scenarios. Therefore, future research needs to prioritize realistic datasets and strategies that reflect the dynamic and evolving nature of network environments to fill the gap between theoretical models and practical implementation.

B. LACK OF EXPLANATION IN PERTURBATION PROCESS

Lack of comprehensive explanations detailing how features are perturbed and the procedures involved in creating adversarial examples is a significant gap in the current research on adversarial attacks against NIDS. This oversight affects the development of efficient countermeasures in addition to restricting the ability of the field to fully understand, replicate, and advance attack techniques. Key areas of concern include:

- 1) **Feature Selection:** Most of current studies [133], [143] fail to explain the adopted criteria for selecting features to perturb. For instance, it is frequently unclear whether the selected features are based on their influence on model predictions, their vulnerability to manipulation, or random selection. Without this information, it is challenging to predict which features might be targeted in real-world scenarios. Moreover, the lack of transparency makes it difficult to evaluate whether the proposed adversarial strategies are practical or merely theoretical constructs.
- 2) **Omission of Perturbation Methodology:** Important information about how features are manipulated are frequently ignored. For example, are the perturbations additive, multiplicative, or based on specific algorithms? How are limitations managed, like making sure no feature is outside of the allowed range? It is impossible to evaluate the realistic and reproducible nature of the perturbation process without these details.
- 3) **Insufficient Evaluation Metrics:** Few studies explain how they evaluate the effectiveness of their attacks beyond classification accuracy reduction. Metrics like semantic validity, transferability, or operational feasibility are often missed. Consequently, it is leaving a gap in understanding how adversarial examples function in real-world deployments.

Understanding how adversarial examples are crafted is essential for designing effective defenses, as it provides insights into the vulnerabilities being exploited. Without this understanding, defenses remain reactive and disconnected from the evolving nature of adversarial strategies.

C. LACK OF ADVERSARIAL DEFENSES

Despite significant advancements in adversarial machine learning, the current state of defenses remains underdeveloped for several reasons:

- **Dependence on Unrealistic Attacks:** The majority of available defenses are designed against unrealistic adversarial attack models, like those that assume complete knowledge of system features or oversimplified threat scenarios. This discrepancy illustrates the necessity of defenses that consider both hypothetical threats and newly emerging real-world situations.
- **Overgeneralized Defenses:** Many defenses are adapted from adversarial machine learning in fields like natural language processing and computer vision. These

techniques, such as adversarial training or input preprocessing, frequently ignore for the structured nature of network traffic, which requires adherence to protocol constraints and time-sensitive patterns.

- **Deficiency in Evaluation Benchmarks:** Adversarial defenses are often evaluated inconsistently. Evaluations are carried out using old datasets or simplified attack scenarios that do not represent changing threats, and metrics are not standardized.

Despite the limited realism of current adversarial attacks, it is essential that adversarial defenses be included in this survey. Since the area of cybersecurity continues to evolve and more complex attacks are expected to emerge, defenses are necessary for NIDS's resilience for potential threats. By analyzing defenses, we not only bring attention to weaknesses in current attack models but also encourage the development of novel adversarial strategies, advancing the field toward practical and reliable solutions. Additionally, the interaction between attacks and defenses promotes iterative development, ensuring that the creation of explainable, scalable, and adaptable NIDS can be guided by lessons learned from hypothetical situations.

D. GOOD PRACTICES

As the volume and complexity of data that must be processed per unit of time continue to increase, systems such as NIDS must utilize techniques like ML for rapid and efficient detection. However, several factors need to be taken into account before implementing ML-based approaches for network intrusion detection:

- **Adversarial Training:** Incorporating adversarial perturbations during the training process of a NIDS inherently increases robustness and resilience against evasion attempts during inference. However the adversarial training should be carefully done by using plausible adversarial data and not just rely on simplistic adversarial generation techniques that would fool the machine learning model but do not represent valid network data. This would allow the model to actually become robust towards attacks that might actually happen, not just overfit on non-plausible adversarial data.
- **Ensemble Learning and Fusion:** In addition to adversarial training, ensemble learning approaches have proven effective against adversarial attacks. Combining multiple diverse architectures, each trained differently, allows them to learn slightly different decision boundaries, improving overall performance and robustness. The independent errors of these models contribute to better performance and increased overall adversarial robustness since not all models are affected similarly by the same adversarial sample. Robustness can be further tuned through different model fusion methods (e.g., voting methods, averaging, and meta-learning), leveraging the strengths of individual models. For ensemble-based NIDS, carefully studying the resilience

of individual models to several adversarial attacks can help tailor model fusion strategies to enhance adversarial robustness, making NIDS less vulnerable to attacks.

- **Robust Feature Engineering:** Incorporating high-level semantic features vs. low-level or raw data in NIDS. Incorporating high-level semantic features rather than low-level or raw data in NIDS generally provides a more robust and resilient foundation for machine learning models. Focusing on semantic understanding and abstraction improves model stability, generalization, and resistance to adversarial attacks, increasing reliability and effectiveness in real-world applications.
- **Input Preprocessing:** Input preprocessing plays a crucial role in enhancing the robustness of machine learning models against adversarial attacks. By applying techniques such as data normalization, feature scaling, noise filtering, and input transformation, preprocessing can reduce the sensitivity of a model to small, malicious perturbations that adversaries introduce to manipulate predictions. These steps help the model focus on the essential patterns within the data rather than being misled by subtle alterations designed to exploit weaknesses. Effective input preprocessing can act as a first line of defense, mitigating the impact of adversarial examples and contributing to a more resilient, reliable system.
- **Anomaly Detection:** Anomaly detection can significantly enhance the adversarial robustness of a machine learning model by identifying deviations from expected or normal behavior, thereby flagging potentially malicious inputs. Since adversarial attacks often involve introducing subtle, crafted perturbations designed to mislead the model, anomaly detection techniques can act as a safeguard by detecting these out-of-distribution inputs that do not align with typical data patterns. Regarding ML-based NIDS, envisioning the detection system as a multi-step approach by having an anomaly filter able to detect these suspicious inputs can be a way to augment the overall performance of the system as a whole.

A variety of studies have proposed detection techniques that specifically target adversarial examples in NIDS. For example, Sarikay et al. [165] presented RAIDS, a powerful autoencoder-based IDS that reconstructs benign traffic patterns to detect anomalies generated by adversarial manipulations. Tuna et al. [166] introduced a closeness and uncertainty-aware strategy that flags adversarial inputs by examining prediction confidence and distributional shifts. Moreover, in [167], the authors employed GANs to discriminate between benign and adversarial traffic, training the model to identify slight deviations. Additionally, in certain NIDS scenarios, Feature Squeezing [168] has been adapted to decrease input variability and reveal abnormal patterns, despite its original design for image-based systems. These studies show how useful detection systems are as an additional

line of defense, especially in situations with limited resources or in real-time.

- **Proper (extensive) Evaluation:** Proper model evaluation is critical for ensuring the robustness of a machine learning model, particularly in the face of adversarial attacks. When it comes to NIDS, particular care must be given to evaluating and stress testing the model (and the IDS in general) in scenarios pertaining to the threats it is envisioned to withstand. As such the system should be thoroughly evaluated on not just standard network data, which in many cases are outdated, but on timely data that includes also corner cases, out-of-distribution data that fall under the overall threat surface envisioned in the threat model of the system.

XI. CONCLUSION AND FUTURE DIRECTIONS

This survey provides a comprehensive and structured analysis of adversarial attacks on ML and DL based NIDS, highlighting the evolving nature of cybersecurity threats. By categorizing ML-NIDS approaches, evaluating their vulnerability to adversarial attacks, and analyzing existing defenses, this study offers a detailed assessment of both theoretical and practical advancements in the field.

Given their practicality and the gaps in existing research on adaptive adversarial techniques, evasion attacks are the main focus of this survey. We reveal significant limitations in current ML-NIDS that adversaries may exploit to evade detection, particularly regarding feature restrictions, semantic validity, and real-time application. In addition, we present practical recommendations to develop reliable ML-NIDS, bridging the gap between academic research and deployment challenges in operational settings.

By identifying new insights, open challenges and future research directions including adversarial example placement, adaptive defenses, and the impact of feature interdependencies; this survey serves as a foundational reference for researchers, cybersecurity practitioners, and organizations aiming to reinforce the resilience of ML-NIDS against adversarial attacks.

REFERENCES

- [1] K. Butler, T. R. Farley, P. McDaniel, and J. Rexford, "A survey of BGP security issues and solutions," *Proc. IEEE*, vol. 98, no. 1, pp. 100–122, Jan. 2010.
- [2] N. Aviram, S. Schinzel, J. Somorovsky, N. Heninger, M. Dankel, J. Steube, L. Valenta, D. Adrian, J. A. Halderman, V. Dukhovni, E. Kasper, S. Cohny, S. Engels, C. Paar, and Y. Shavitt, "DROWN: Breaking TLS using SSLv2," in *Proc. 25th USENIX Secur. Symp. (USENIX Secur.)*, 2016, pp. 689–706.
- [3] G. Xiong, J. Tong, Y. Xu, H. Yu, and Y. Zhao, "A survey of network attacks based on protocol vulnerabilities," in *Proc. Asia-Pacific Web Conf. Technol. Appl.*, 2014, pp. 246–257.
- [4] M. Conti, F. De Gaspari, and L. V. Mancini, "A novel stealthy attack to gather SDN configuration-information," *IEEE Trans. Emerg. Topics Comput.*, vol. 8, no. 2, pp. 328–340, Apr. 2020.
- [5] Z. Durumeric, F. Li, J. Kasten, J. Amann, J. Beekman, M. Payer, N. Weaver, D. Adrian, V. Paxson, M. Bailey, and J. A. Halderman, "The matter of heartbleed," in *Proc. Conf. Internet Meas. Conf.*, Nov. 2014, pp. 475–488.
- [6] M. Antonakakis, T. April, M. Bailey, M. Bernhard, E. Bursztein, J. Cochran, Z. Durumeric, J. A. Halderman, L. Invernizzi, M. Kallitsis, D. Kumar, C. Lever, Z. Ma, J. Mason, D. Menscher, C. Seaman, N. Sullivan, K. Thomas, and Y. Zhou, "Understanding the Mirai botnet," in *Proc. 26th USENIX Secur. Symp. (USENIX Secur.)*, 2017, pp. 1093–1110.
- [7] M. G. Gouda and A. X. Liu, "Structured firewall design," *Comput. Netw.*, vol. 51, no. 4, pp. 1106–1120, Mar. 2007.
- [8] P. Matta, M. Arora, and D. Sharma, "A comparative survey on data encryption techniques: Big data perspective," *Mater. Today, Proc.*, vol. 46, pp. 11035–11039, Jan. 2021.
- [9] D. Gibert, C. Mateu, J. Planes, and J. Marques-Silva, "Auditing static machine learning anti-malware tools against metamorphic attacks," *Comput. Secur.*, vol. 102, Mar. 2021, Art. no. 102159.
- [10] P. Theron, A. Kott, M. Drašar, K. Rzađar, B. LeBlanc, M. Pihelgas, L. Mancini, and F. de Gaspari, "Reference architecture of an autonomous agent for cyber defense of complex military systems," in *Adaptive Autonomous Secure Cyber Systems*, 2020, pp. 1–21.
- [11] F. De Gaspari, S. Jajodia, L. V. Mancini, and G. Pagnotta, *Towards Intelligent Cyber Deception Systems*. Cham, Switzerland: Springer, 2019.
- [12] S. Jajodia, A. K. Ghosh, V. Swarup, C. Wang, and X. S. Wang, *Moving Target Defense: Creating Asymmetric Uncertainty for Cyber Threats*, vol. 54. Cham, Switzerland: Springer, 2011.
- [13] G. Pagnotta, F. De Gaspari, D. Hitaj, M. Andreolini, M. Colajanni, and L. V. Mancini, "DOLOS: A novel architecture for moving target defense," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 5890–5905, 2023.
- [14] M. Ozkan-Okay, R. Samet, Ö. Aslan, and D. Gupta, "A comprehensive systematic literature review on intrusion detection systems," *IEEE Access*, vol. 9, pp. 157727–157760, 2021.
- [15] O. Can and O. K. Sahingoz, "A survey of intrusion detection systems in wireless sensor networks," in *Proc. 6th Int. Conf. Modeling, Simulation, Appl. Optim. (ICMSAO)*, May 2015, pp. 1–6.
- [16] D. Chou and M. Jiang, "A survey on data-driven network intrusion detection," *ACM Comput. Surveys*, vol. 54, no. 9, pp. 1–36, Dec. 2022.
- [17] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
- [18] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1800–1807.
- [19] Y. Kim, "Convolutional neural networks for sentence classification," 2014, *arXiv:1408.5882*.
- [20] G. Pagnotta, D. Hitaj, F. De Gaspari, and L. V. Mancini, "Pass-Flow: Guessing passwords with generative flows," in *Proc. 52nd Annu. IEEE/IFIP Int. Conf. Dependable Syst. Netw. (DSN)*, Jun. 2022, pp. 251–262.
- [21] A. Graves, A.-R. Mohamed, and G. Hinton, "Speech recognition with deep recurrent neural networks," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, May 2013, pp. 6645–6649.
- [22] G. Hinton, L. Deng, D. Yu, G. E. Dahl, A.-R. Mohamed, N. Jaitly, A. Senior, V. Vanhoucke, P. Nguyen, T. N. Sainath, and B. Kingsbury, "Deep neural networks for acoustic modeling in speech recognition: The shared views of four research groups," *IEEE Signal Process. Mag.*, vol. 29, no. 6, pp. 82–97, Nov. 2012.
- [23] B. Hitaj, P. Gasti, G. Ateniese, and F. Pérez-Cruz, "PassGAN: A deep learning approach for password guessing," in *Proc. Int. Conf. Appl. Cryptogr. Netw. Secur.*, 2019, pp. 217–237.
- [24] T. Karras, S. Laine, and T. Aila, "A style-based generator architecture for generative adversarial networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4401–4410.
- [25] D. Hitaj, G. Pagnotta, B. Hitaj, L. V. Mancini, and F. Pérez-Cruz, "MaleficNet: Hiding malware into deep neural networks using spread-spectrum channel coding," in *Proc. Eur. Symp. Res. Comput. Secur.*, 2022, pp. 425–444.
- [26] D. Hitaj, G. Pagnotta, B. Hitaj, F. Pérez-Cruz, and L. V. Mancini, "FedComm: Federated learning as a medium for covert communication," *IEEE Trans. Dependable Secure Comput.*, vol. 21, no. 4, pp. 1695–1707, Jul. 2024.
- [27] F. D. Gaspari, D. Hitaj, G. Pagnotta, L. D. Carli, and L. V. Mancini, "Reliable detection of compressed and encrypted data," *Neural Comput. Appl.*, vol. 34, pp. 20379–20393, Jan. 2022.

- [28] F. D. Gaspari, D. Hitaj, G. Pagnotta, L. D. Carli, and L. V. Mancini, "EnCoD: Distinguishing compressed and encrypted file fragments," in *Proc. 14th Int. Conf. Netw. Syst. Secur.*, 2020, pp. 42–62.
- [29] W. Marx. (2021). *How Valencia Crushed COVID With AI*. [Online]. Available: <https://www.wired.co.uk/article/valencia-ai-covid-data>
- [30] M. Á. Lozano, Ö. Garibo-I-Orts, E. Piñol, M. Rebollo, K. Polotskaya, M. Á. García-March, J. A. Conejero, F. Escolano, and N. Oliver, "Open data science to fight COVID-19: Winning the 500k XPRIZE pandemic response challenge," in *Proc. Joint Eur. Conf. Mach. Learn. Knowl. Discovery Databases*, 2021, pp. 384–399.
- [31] M. Barreno, B. Nelson, A. D. Joseph, and J. D. Tygar, "The security of machine learning," *Mach. Learning*, vol. 81, no. 2, pp. 121–148, 2010.
- [32] A. Michel, S. K. Jha, and R. Ewet, "A survey on the vulnerability of deep neural networks against adversarial attacks," *Prog. Artif. Intell.*, vol. 11, no. 2, pp. 1–11, Jun. 2022.
- [33] Z. Tian, L. Cui, J. Liang, and S. Yu, "A comprehensive survey on poisoning attacks and countermeasures in machine learning," *ACM Comput. Surveys*, vol. 55, no. 8, pp. 1–35, Aug. 2023.
- [34] F. D. Gaspari, D. Hitaj, and L. V. Mancini, "Have you poisoned my data? Defending neural networks against data poisoning," in *Proc. Eur. Symp. Res. Comput. Secur.*, 2024, pp. 85–104.
- [35] I. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," in *Proc. Int. Conf. Learn. Represent.*, 2014.
- [36] J. Zhang and C. Li, "Adversarial examples: Opportunities and challenges," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 7, pp. 2578–2593, Jul. 2020.
- [37] Z. Lin, Y. Shi, and Z. Xue, "IDSGAN: Generative adversarial networks for attack generation against intrusion detection," in *Proc. 26th Pacific-Asia Conf. Adv. Knowl. Discovery Data Mining*, Chengdu, China, 2022, pp. 79–91.
- [38] M. Rigaki and S. Garcia, "Bringing a GAN to a knife-fight: Adapting malware communication to avoid detection," in *Proc. IEEE Secur. Privacy Workshops (SPW)*, May 2018, pp. 70–75.
- [39] D. Wu, B. Fang, J. Wang, Q. Liu, and X. Cui, "Evading machine learning botnet detection models via deep reinforcement learning," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2019, pp. 1–6.
- [40] F. D. Gaspari, D. Hitaj, G. Pagnotta, L. D. Carli, and L. V. Mancini, "The naked sun: Malicious cooperation between benign-looking processes," in *Proc. Int. Conf. Appl. Cryptogr. Netw. Secur.*, 2020, pp. 254–274.
- [41] D. Shu, N. O. Leslie, C. A. Kamhoua, and C. S. Tucker, "Generative adversarial attacks against intrusion detection systems using active learning," in *Proc. 2nd ACM Workshop Wireless Secur. Mach. Learn.*, Jul. 2020, pp. 1–6.
- [42] B.-E. Zolbayar, R. Sheatsley, P. McDaniel, M. J. Weisman, S. Zhu, S. Zhu, and S. Krishnamurthy, "Generating practical adversarial network traffic flows using NIDSGAN," 2022, *arXiv:2203.06694*.
- [43] I. Debicha, T. Debatty, J.-M. Dricot, and W. Mees, "Adversarial training for deep learning-based intrusion detection systems," 2021, *arXiv:2104.09852*.
- [44] A. L. Buczak and E. Guven, "A survey of data mining and machine learning methods for cyber security intrusion detection," *IEEE Commun. Surveys Tuts.*, vol. 18, no. 2, pp. 1153–1176, 2nd Quart., 2016.
- [45] N. Chaabouni, M. Mosbah, A. Zemmar, C. Sauvignac, and P. Faruki, "Network intrusion detection for IoT security based on learning techniques," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 3, pp. 2671–2701, 3rd Quart., 2019.
- [46] Z. Ahmad, A. Shahid Khan, C. Wai Shiang, J. Abdullah, and F. Ahmad, "Network intrusion detection system: A systematic study of machine learning and deep learning approaches," *Trans. Emerg. Telecommun. Technol.*, vol. 32, no. 1, p. 4150, Jan. 2021.
- [47] A. Drewek-Ossowicka, M. Pietrolaj, and J. Rumiński, "A survey of neural networks usage for intrusion detection systems," *J. Ambient Intell. Humanized Comput.*, vol. 12, no. 1, pp. 497–514, Jan. 2021.
- [48] A. Thakkar and R. Lohiya, "A survey on intrusion detection system: Feature selection, model, performance measures, application perspective, challenges, and future research directions," *Artif. Intell. Rev.*, vol. 55, no. 1, pp. 453–563, Jan. 2022.
- [49] T. R. Devi and S. Badugu, "A review on network intrusion detection system using machine learning," in *Proc. Adv. Decis. Sci.*, 2019, pp. 598–607.
- [50] S. Zhou, C. Liu, D. Ye, T. Zhu, W. Zhou, and P. S. Yu, "Adversarial attacks and defenses in deep learning: From a perspective of cybersecurity," *ACM Comput. Surveys*, vol. 55, no. 8, pp. 1–39, Aug. 2023.
- [51] P. Xiong, M. Tegegn, J. S. Sarin, S. Pal, and J. Rubin, "It is all about data: A survey on the effects of data on adversarial robustness," *ACM Comput. Surveys*, vol. 56, no. 7, pp. 1–41, Jul. 2024.
- [52] J. Gardiner and S. Nagaraja, "On the security of machine learning in malware C&C detection: A survey," *ACM Comput. Surv.*, vol. 49, no. 3, pp. 1–39, 2016.
- [53] D. Park and B. Yener, "A survey on practical adversarial examples for malware classifiers," in *Proc. Reversing Offensive-Oriented Trends Symp.*, Nov. 2020, pp. 23–35.
- [54] I. Rosenberg, A. Shabtai, Y. Elovici, and L. Rokach, "Adversarial machine learning attacks and defense methods in the cyber security domain," *ACM Comput. Surveys*, vol. 54, no. 5, pp. 1–36, Jun. 2022.
- [55] N. Akhtar, A. Mian, N. Kardan, and M. Shah, "Advances in adversarial attacks and defenses in computer vision: A survey," *IEEE Access*, vol. 9, pp. 155161–155196, 2021.
- [56] J. Lan, R. Zhang, Z. Yan, J. Wang, Y. Chen, and R. Hou, "Adversarial attacks and defenses in speaker recognition systems: A survey," *J. Syst. Archit.*, vol. 127, Jun. 2022, Art. no. 102526.
- [57] O. Ibitoye, R. Abou-Khamis, M. E. Shehaby, A. Matrawy, and M. O. Shafiq, "The threat of adversarial attacks on machine learning in network security—A survey," 2019, *arXiv:1911.02621*.
- [58] F. O. Olowononi, D. B. Rawat, and C. Liu, "Resilient machine learning for networked cyber physical systems: A survey for machine learning security to securing machine learning for CPS," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 1, pp. 524–552, 1st Quart., 2021.
- [59] N. Martins, J. M. Cruz, T. Cruz, and P. Henriques Abreu, "Adversarial machine learning applied to intrusion and malware scenarios: A systematic review," *IEEE Access*, vol. 8, pp. 35403–35419, 2020.
- [60] A. Alotaibi and M. A. Rassam, "Adversarial machine learning attacks against intrusion detection systems: A survey on strategies and defense," *Future Internet*, vol. 15, no. 2, p. 62, Jan. 2023.
- [61] A. Pinto, L.-C. Herrera, Y. Donoso, and J. A. Gutierrez, "Survey on intrusion detection systems based on machine learning techniques for the protection of critical infrastructure," *Sensors*, vol. 23, no. 5, p. 2415, Feb. 2023.
- [62] K. He, D. D. Kim, and M. R. Asghar, "Adversarial machine learning for network intrusion detection systems: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 1, pp. 538–566, 1st Quart., 2023.
- [63] X. Ma, B. Li, Y. Wang, S. M. Erfani, S. Wijewickrema, G. Schoenebeck, D. Song, M. E. Houle, and J. Bailey, "Characterizing adversarial subspaces using local intrinsic dimensionality," in *Proc. Int. Conf. Learn. Represent.*, 2018, pp. 1–15.
- [64] A. Aldweesh, A. Derhab, and A. Z. Emam, "Deep learning approaches for anomaly-based intrusion detection systems: A survey, taxonomy, and open issues," *Knowl.-Based Syst.*, vol. 189, Feb. 2020, Art. no. 105124.
- [65] N. Hubballi and V. Suryanarayanan, "False alarm minimization techniques in signature-based intrusion detection systems: A survey," *Comput. Commun.*, vol. 49, pp. 1–17, Aug. 2014.
- [66] T. Saranya, S. Sridevi, C. Deisy, T. D. Chung, and M. K. A. A. Khan, "Performance analysis of machine learning algorithms in intrusion detection system: A review," *Proc. Comput. Sci.*, vol. 171, pp. 1251–1260, Jan. 2020.
- [67] G. Chiron, F. Arrestier, and A. M. Awal, "Fast end-to-end deep learning identity document detection, classification and cropping," in *Proc. 16th Int. Conf. Document Anal. Recognit.*, 2021, pp. 333–347.
- [68] G. Marín, P. Casas, and G. Capdehourat, "RawPower: Deep learning based anomaly detection from raw network traffic measurements," in *Proc. ACM SIGCOMM Conf. Posters Demos*, Aug. 2018, pp. 75–77.
- [69] M. Emani et al., "A comprehensive evaluation of novel AI accelerators for deep learning workloads," in *Proc. IEEE/ACM Int. Workshop Perform. Modeling, Benchmarking Simulation High Perform. Comput. Syst. (PMBS)*, Nov. 2022, pp. 13–25.
- [70] M. A. Merzouk, F. Cuppens, N. Boulahia-Cuppens, and R. Yaich, "Investigating the practicality of adversarial evasion attacks on network intrusion detection," *Ann. Telecommun.*, vol. 77, nos. 11–12, pp. 763–775, Dec. 2022.
- [71] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus, "Intriguing properties of neural networks," in *Proc. Int. Conf. Learn. Represent.*, 2014.

- [72] A. Kurakin, I. J. Goodfellow, and S. Bengio, "Adversarial examples in the physical world," in *Proc. Int. Conf. Learn. Represent.*, 2017, pp. 1–7.
- [73] S.-M. Moosavi-Dezfooli, A. Fawzi, and P. Frossard, "DeepFool: A simple and accurate method to fool deep neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2574–2582.
- [74] N. Carlini and D. Wagner, "Towards evaluating the robustness of neural networks," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2017, pp. 39–57.
- [75] P. Linardatos, V. Papastefanopoulos, and S. Kotsiantis, "Explainable AI: A review of machine learning interpretability methods," *Entropy*, vol. 23, no. 1, p. 18, Dec. 2020.
- [76] I. Goodfellow, "NIPS 2016 tutorial: Generative adversarial networks," 2017, *arXiv:1701.00160*.
- [77] P.-Y. Chen, H. Zhang, Y. Sharma, J. Yi, and C.-J. Hsieh, "ZOO: Zeroth order optimization based black-box attacks to deep neural networks without training substitute models," in *Proc. 10th ACM Workshop Artif. Intell. Secur.*, Nov. 2017, pp. 15–26.
- [78] M. Zhou, J. Wu, Y. Liu, S. Liu, and C. Zhu, "DaST: Data-free substitute training for adversarial attacks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 234–243.
- [79] X. Yang, Y. Lyu, T. Tian, Y. Liu, Y. Liu, and X. Zhang, "Rumor detection on social media with graph structured adversarial learning," in *Proc. 29th Int. Joint Conf. Artif. Intell.*, Jul. 2020, pp. 1417–1423.
- [80] K. Eykholt, I. Evtimov, E. Fernandes, B. Li, A. Rahmati, C. Xiao, A. Prakash, T. Kohno, and D. Song, "Robust physical-world attacks on deep learning visual classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1625–1634.
- [81] Y. Liu, S. Ma, Y. Aafer, W.-C. Lee, J. Zhai, W. Wang, and X. Zhang, "Trojaning attack on neural networks," in *Proc. Netw. Distrib. Syst. Secur. Symp.*, 2018.
- [82] N. Carlini and D. Wagner, "Adversarial examples are not easily detected: Bypassing ten detection methods," in *Proc. 10th ACM Workshop Artif. Intell. Secur.*, Nov. 2017, pp. 3–14.
- [83] N. Carlini and D. Wagner, "MagNet and 'efficient defenses against adversarial attacks' are not robust to adversarial examples," 2017, *arXiv:1711.08478*.
- [84] A. Athalye, N. Carlini, and D. T. Wagner, "Obfuscated gradients give a false sense of security: Circumventing defenses to adversarial examples," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 274–283.
- [85] N. Carlini, A. Athalye, N. Papernot, W. Brendel, J. Rauber, D. Tsipras, I. Goodfellow, A. Madry, and A. Kurakin, "On evaluating adversarial robustness," 2019, *arXiv:1902.06705*.
- [86] F. Tramèr, N. Carlini, W. Brendel, and A. Madry, "On adaptive attacks to adversarial example defenses," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 1633–1645.
- [87] J. Hendrik Metzen, T. Genewein, V. Fischer, and B. Bischoff, "On detecting adversarial perturbations," 2017, *arXiv:1702.04267*.
- [88] Z. T. M. Ali, A. Mohammed, and I. Ahmad, "Evaluating adversarial robustness of secret key-based defenses," *IEEE Access*, vol. 10, pp. 34872–34882, 2022.
- [89] Y. Chen, M. Zhang, J. Li, and X. Kuang, "Adversarial attacks and defenses in image classification: A practical perspective," in *Proc. 7th Int. Conf. Image, Vis. Comput. (ICIVC)*, Jul. 2022, pp. 424–430.
- [90] D. Stutz, M. Hein, and B. Schiele, "Disentangling adversarial robustness and generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6969–6980.
- [91] I. Tolstikhin, O. Bousquet, S. Gelly, and B. Schölkopf, "Wasserstein autoencoders," in *Proc. 6th Int. Conf. Learn. Represent. (ICLR)*, 2018.
- [92] T. Pang, K. Xu, C. Du, N. Chen, and J. Zhu, "Improving adversarial robustness via promoting ensemble diversity," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 4970–4979.
- [93] G. Apruzzese, M. Andreolini, L. Ferretti, M. Marchetti, and M. Colajanni, "Modeling realistic adversarial attacks against network intrusion detection systems," *Digit. Threats, Res. Pract.*, vol. 3, no. 3, pp. 1–19, Sep. 2022.
- [94] H. Mohammadian, A. H. Lashkari, and A. A. Ghorbani, "Poisoning and evasion: Deep learning-based NIDS under adversarial attacks," in *Proc. 21st Annu. Int. Conf. Privacy, Secur. Trust (PST)*, Aug. 2024, pp. 1–9.
- [95] T. Das, R. M. Shukla, and S. Sengupta, "Poisoning the well: Adversarial poisoning on ML-based software-defined network intrusion detection systems," *IEEE Trans. Netw. Sci. Eng.*, vol. 12, no. 1, pp. 252–262, Jan. 2025.
- [96] S. Alahmed, Q. Alasad, J.-S. Yuan, and M. Alawad, "Impacting robustness in deep learning-based NIDS through poisoning attacks," *Algorithms*, vol. 17, no. 4, p. 155, Apr. 2024.
- [97] G. Xia, J. Chen, C. Yu, and J. Ma, "Poisoning attacks in federated learning: A survey," *IEEE Access*, vol. 11, pp. 10708–10722, 2023.
- [98] P. Chen, X. Du, Z. Lu, and H. Chai, "Universal adversarial backdoor attacks to fool vertical federated learning," *Comput. Secur.*, vol. 137, Feb. 2024, Art. no. 103601.
- [99] Z. Chkibene, A. Erbad, R. Hamila, A. Mohamed, M. Guizani, and M. Hamdi, "TIDCS: A dynamic intrusion detection and classification system based feature selection," *IEEE Access*, vol. 8, pp. 95864–95877, 2020.
- [100] B. A. AlAhmadi, E. Mariconti, R. Spolaor, G. Stringhini, and I. Martinovic, "BOTection: Bot detection by building Markov chain models of bots network behavior," in *Proc. 15th ACM Asia Conf. Comput. Commun. Secur.*, Oct. 2020, pp. 652–664.
- [101] M. Nunes, P. Burnap, P. Reinecke, and K. Lloyd, "Bane or boon: Measuring the effect of evasive malware on system call classifiers," *J. Inf. Secur. Appl.*, vol. 67, Jun. 2022, Art. no. 103202.
- [102] S. Y. Mehr and B. Ramamurthy, "An SVM based DDoS attack detection method for ryu SDN controller," in *Proc. 15th Int. Conf. Emerg. Netw. EXperiments Technol.*, Dec. 2019, pp. 72–73.
- [103] Y. Zhou, M. Han, L. Liu, J. S. He, and Y. Wang, "Deep learning approach for cyberattack detection," in *Proc. IEEE Conf. Comput. Commun. Workshops (INFOCOM WKSHPS)*, Apr. 2018, pp. 262–267.
- [104] S. Manimurugan, S. Al-Mutairi, M. M. Aborokbah, N. Chilamkurti, S. Ganesan, and R. Patan, "Effective attack detection in Internet of Medical Things smart environment using a deep belief neural network," *IEEE Access*, vol. 8, pp. 77396–77404, 2020.
- [105] M. Z. Alom and T. M. Taha, "Network intrusion detection for cyber security on neuromorphic computing system," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, May 2017, pp. 3830–3837.
- [106] M. Pawlicki, M. Choraś, and R. Kozik, "Defending network intrusion detection systems against adversarial evasion attacks," *Future Gener. Comput. Syst.*, vol. 110, pp. 148–154, Sep. 2020.
- [107] F. Laghrissi, S. Douzi, K. Douzi, and B. Hssina, "Intrusion detection systems using long short-term memory (LSTM)," *J. Big Data*, vol. 8, no. 1, p. 65, Dec. 2021.
- [108] X. Kan, Y. Fan, Z. Fang, L. Cao, N. N. Xiong, D. Yang, and X. Li, "A novel IoT network intrusion detection approach based on adaptive particle swarm optimization convolutional neural network," *Inf. Sci.*, vol. 568, pp. 147–162, Aug. 2021.
- [109] A. Thakkar and R. Lohiya, "Fusion of statistical importance for feature selection in deep neural network-based intrusion detection system," *Inf. Fusion*, vol. 90, pp. 353–363, Feb. 2023.
- [110] M. Piskozub, F. De Gaspari, F. Barr-Smith, L. Mancini, and I. Martinovic, "MalPhase: Fine-grained malware detection using network flow data," in *Proc. ACM Asia Conf. Comput. Commun. Secur.*, May 2021, pp. 774–786.
- [111] Y. Zhong, W. Chen, Z. Wang, Y. Chen, K. Wang, Y. Li, X. Yin, X. Shi, J. Yang, and K. Li, "HELAD: A novel network anomaly detection model based on heterogeneous ensemble learning," *Comput. Netw.*, vol. 169, Mar. 2020, Art. no. 107049.
- [112] Y. Mirsky, T. Doitshman, Y. Elovici, and A. Shabtai, "Kitsune: An ensemble of autoencoders for online network intrusion detection," 2018, *arXiv:1802.09089*.
- [113] B. A. Tama, M. Comuzzi, and K.-H. Rhee, "TSE-IDS: A two-stage classifier ensemble for intelligent anomaly-based intrusion detection system," *IEEE Access*, vol. 7, pp. 94497–94507, 2019.
- [114] Y. Zhou, G. Cheng, S. Jiang, and M. Dai, "Building an efficient intrusion detection system based on feature selection and ensemble classifier," *Comput. Netw.*, vol. 174, Jun. 2020, Art. no. 107247.
- [115] D. Xibin, Z. Yu, W. Cao, Y. Shi, and Q. Ma, "A survey on ensemble learning," *Frontiers Comput. Sci.*, vol. 14, no. 2, pp. 241–258, 2019.
- [116] J. Gu, L. Wang, H. Wang, and S. Wang, "A novel approach to intrusion detection using SVM ensemble with feature augmentation," *Comput. Secur.*, vol. 86, pp. 53–62, Sep. 2019.
- [117] N. Marir, H. Wang, G. Feng, B. Li, and M. Jia, "Distributed abnormal behavior detection approach based on deep belief network and ensemble SVM using spark," *IEEE Access*, vol. 6, pp. 59657–59671, 2018.
- [118] E.-U.-H. Qazi, M. Imran, N. Haider, M. Shoaib, and I. Razzak, "An intelligent and efficient network intrusion detection system using deep learning," *Comput. Electr. Eng.*, vol. 99, Apr. 2022, Art. no. 107764.

- [119] V. Ravi, R. Chaganti, and M. Alazab, "Recurrent deep learning-based feature fusion ensemble meta-classifier approach for intelligent network intrusion detection system," *Comput. Electr. Eng.*, vol. 102, Sep. 2022, Art. no. 108156.
- [120] A. Afianian, S. Niksefat, B. Sadeghiyan, and D. Baptiste, "Malware dynamic analysis evasion techniques: A survey," *ACM Comput. Surveys*, vol. 52, no. 6, pp. 1–28, Nov. 2020.
- [121] O. Arreche, I. Bibers, and M. Abdallah, "A two-level ensemble learning framework for enhancing network intrusion detection systems," *IEEE Access*, vol. 12, pp. 83830–83857, 2024.
- [122] H. Nandanwar and R. Katarya, "Deep learning enabled intrusion detection system for industrial IoT environment," *Expert Syst. Appl.*, vol. 249, Sep. 2024, Art. no. 123808.
- [123] T. Hasan, J. Malik, I. Bibi, W. U. Khan, F. N. Al-Wesabi, K. Dev, and G. Huang, "Securing industrial Internet of Things against botnet attacks using hybrid deep learning approach," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 5, pp. 2952–2963, Sep. 2023.
- [124] A. Nazir, J. He, N. Zhu, S. S. Qureshi, S. U. Qureshi, F. Ullah, A. Wajahat, and M. S. Pathan, "A deep learning-based novel hybrid CNN-LSTM architecture for efficient detection of threats in the IoT ecosystem," *Ain Shams Eng. J.*, vol. 15, no. 7, Jul. 2024, Art. no. 102777.
- [125] A. L. Yaser, H. M. Mousa, and M. Hussein, "Improved DDoS detection utilizing deep neural networks and feedforward neural networks as autoencoder," *Future Internet*, vol. 14, no. 8, p. 240, Aug. 2022.
- [126] Y. Meidan, M. Bohadana, Y. Mathov, Y. Mirsky, A. Shabtai, D. Breitenbacher, and Y. Elovici, "N-BaIoT—Network-based detection of IoT botnet attacks using deep autoencoders," *IEEE Pervasive Comput.*, vol. 17, no. 3, pp. 12–22, Sep. 2018.
- [127] H. Mohammadian, A. A. Ghorbani, and A. H. Lashkari, "A gradient-based approach for adversarial attack on deep learning-based network intrusion detection systems," *Appl. Soft Comput.*, vol. 137, Apr. 2023, Art. no. 110173.
- [128] N. Papernot, P. McDaniel, S. Jha, M. Fredrikson, Z. B. Celik, and A. Swami, "The limitations of deep learning in adversarial settings," in *Proc. IEEE Eur. Symp. Secur. Privacy (EuroS&P)*, Mar. 2016, pp. 372–387.
- [129] M. A. Ayub, W. A. Johnson, D. A. Talbert, and A. Siraj, "Model evasion attack on intrusion detection systems using adversarial machine learning," in *Proc. 54th Annu. Conf. Inf. Sci. Syst. (CISS)*, Mar. 2020, pp. 1–6.
- [130] M. Pawlicki, R. Kozik, and M. Choraś, "Artificial neural network hyperparameter optimisation for network intrusion detection," in *Proc. 15th Int. Conf. Intell. Comput. Theories Appl.*, 2019, pp. 749–760.
- [131] A. Mądry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *Proc. Int. Conf. Learn. Represent.*, 2017, pp. 1–17.
- [132] M. J. Hashemi, G. Cusack, and E. Keller, "Towards evaluation of NIDSs in adversarial setting," in *Proc. 3rd ACM CoNEXT Workshop Big Data, Mach. Learn. Artif. Intell. Data Commun. Netw.*, Dec. 2019, pp. 14–21.
- [133] E. Anthi, L. Williams, M. Rhode, P. Burnap, and A. Wedgbury, "Adversarial attacks on machine learning cybersecurity defences in industrial control systems," *J. Inf. Secur. Appl.*, vol. 58, May 2021, Art. no. 102717.
- [134] E. Alhajjar, P. Maxwell, and N. Bastian, "Adversarial machine learning in network intrusion detection systems," *Expert Syst. Appl.*, vol. 186, Dec. 2021, Art. no. 115782.
- [135] C. Zhang, X. Costa-Perez, and P. Patras, "Adversarial attacks against deep learning-based network intrusion detection systems and defense mechanisms," *IEEE/ACM Trans. Netw.*, vol. 30, no. 3, pp. 1294–1311, Jun. 2022.
- [136] A. Ilyas, L. Engstrom, A. Athalye, and J. Lin, "Black-box adversarial attacks with limited queries and information," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 2137–2146.
- [137] W. Brendel, J. Rauber, and M. Bethge, "Decision-based adversarial attacks: Reliable attacks against black-box machine learning models," 2017, *arXiv:1712.04248*.
- [138] J. Chen, M. I. Jordan, and M. J. Wainwright, "HopSkipJumpAttack: A query-efficient decision-based attack," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2020, pp. 1277–1294.
- [139] L. Schott, J. Rauber, M. Bethge, and W. Brendel, "Towards the first adversarially robust neural network model on MNIST," 2018, *arXiv:1805.09190*.
- [140] S. Liu, J. Sun, and J. Li, "Query-efficient hard-label black-box attacks using biased sampling," in *Proc. Chin. Autom. Congr. (CAC)*, Nov. 2020, pp. 3872–3877.
- [141] X. Peng, W. Huang, and Z. Shi, "Adversarial attack against DoS intrusion detection: An improved boundary-based method," in *Proc. IEEE 31st Int. Conf. Tools Artif. Intell. (ICTAI)*, Nov. 2019, pp. 1288–1295.
- [142] Y. Sharon, D. Berend, Y. Liu, A. Shabtai, and Y. Elovici, "TANTRA: Timing-based adversarial network traffic reshaping attack," *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 3225–3237, 2022.
- [143] I. Debicha, B. Cochez, T. Kenaza, T. Debatty, J.-M. Dricot, and W. Mees, "Adv-bot: Realistic adversarial botnet attacks against network intrusion detection systems," *Comput. Secur.*, vol. 129, Jun. 2023, Art. no. 103176.
- [144] J. Vitorino, N. Oliveira, and I. Praça, "Adaptive perturbation patterns: Realistic adversarial learning for robust intrusion detection," *Future Internet*, vol. 14, no. 4, p. 108, Mar. 2022.
- [145] R. Sheatsley, N. Papernot, M. J. Weisman, G. Verma, and P. McDaniel, "Adversarial examples for network intrusion detection systems," *J. Comput. Secur.*, vol. 30, no. 5, pp. 727–752, Oct. 2022.
- [146] H. Jmila and M. I. Khedher, "Adversarial machine learning for network intrusion detection: A comparative study," *Comput. Netw.*, vol. 214, Sep. 2022, Art. no. 109073.
- [147] D. Han, Z. Wang, Y. Zhong, W. Chen, J. Yang, S. Lu, X. Shi, and X. Yin, "Evaluating and improving adversarial robustness of machine learning-based network intrusion detectors," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 8, pp. 2632–2647, Aug. 2021.
- [148] H. A. Akarte and D. K. Yadav, "Packet processing and data plane program verification: A survey with tools, techniques, and challenges," *Int. J. Commun. Syst.*, vol. 36, no. 14, p. 5554, Sep. 2023.
- [149] J. Wen, G. Xu, Z. Tang, W. Wang, L. Fei, and Y. Xu, "Graph regularized and feature aware matrix factorization for robust incomplete multi-view clustering," *IEEE Trans. Circuits Syst. for Video Technol.*, vol. 34, no. 5, pp. 3728–3741, May 2024.
- [150] J. Xie, G. Li, L. Zhang, G. Cheng, K. Zhang, and M. Bai, "Texture feature-aware consistency for semi-supervised honeycomb lung lesion segmentation," *Expert Syst. Appl.*, vol. 258, Dec. 2024, Art. no. 125119.
- [151] Y. Shi, K. Davaslioglu, and Y. E. Sagduyu, "Generative adversarial network in the air: Deep adversarial learning for wireless signal spoofing," *IEEE Trans. Cognit. Commun. Netw.*, vol. 7, no. 1, pp. 294–303, Mar. 2021.
- [152] Y. Jiang and G. Yin, "Attention-enhanced one-shot attack against single object tracking for unmanned aerial vehicle remote sensing images," *Remote Sens.*, vol. 15, no. 18, p. 4514, Sep. 2023.
- [153] Z. Wei, J. Chen, X. Wei, L. Jiang, T. Chua, F. Zhou, and Y. Jiang, "Heuristic black-box adversarial attacks on video recognition models," in *Proc. AAAI Conf. Artif. Intell.*, 2019, pp. 12338–12345.
- [154] S. Alahmed, Q. Alasad, M. M. Hammood, J.-S. Yuan, and M. Alawad, "Mitigation of black-box attacks on intrusion detection systems-based ML," *Computers*, vol. 11, no. 7, p. 115, Jul. 2022.
- [155] A. Venturi, M. Ferrari, M. Marchetti, and M. Colajanni, "ARGANIDS: A novel network intrusion detection system based on adversarially regularized graph autoencoder," in *Proc. 38th ACM/SIGAPP Symp. Appl. Comput.*, Mar. 2023, pp. 1540–1548.
- [156] S. Li, J. Wang, Y. Wang, G. Zhou, and Y. Zhao, "EIFDAA: Evaluation of an IDS with function-discarding adversarial attacks in the IIoT," *Heliyon*, vol. 9, no. 2, Feb. 2023, Art. no. e13520.
- [157] H. Jiang, J. Lin, and H. Kang, "FGMD: A robust detector against adversarial attacks in the IoT network," *Future Gener. Comput. Syst.*, vol. 132, pp. 194–210, Jul. 2022.
- [158] C. Zhang, X. Costa-Perez, and P. Patras, "Tiki-taka: Attacking and defending deep learning-based intrusion detection systems," in *Proc. ACM SIGSAC Conf. Cloud Comput. Secur. Workshop*, Nov. 2020, pp. 27–39.
- [159] A. McCarthy, E. Ghadafi, P. Andriotis, and P. Legg, "Defending against adversarial machine learning attacks using hierarchical learning: A case study on network traffic attack classification," *J. Inf. Secur. Appl.*, vol. 72, Feb. 2023, Art. no. 103398.
- [160] N. Wang, Y. Chen, Y. Xiao, Y. Hu, W. Lou, and Y. T. Hou, "MANDA: On adversarial example detection for network intrusion detection system," *IEEE Trans. Dependable Secure Comput.*, vol. 20, no. 2, pp. 1139–1153, Mar. 2023.
- [161] T. Tanay and L. Griffin, "A boundary tilting perspective on the phenomenon of adversarial examples," 2016, *arXiv:1608.07690*.

- [162] A. Jalal, A. Ilyas, C. Daskalakis, and A. G. Dimakis, "The robust manifold defense: Adversarial training using generative models," 2017, *arXiv:1712.09196*.
- [163] P. Samangouei, M. Kabkab, and R. Chellappa, "Defense-GAN: Protecting classifiers against adversarial attacks using generative models," 2018, *arXiv:1805.06605*.
- [164] J. Gilmer, L. Metz, F. Faghri, S. S. Schoenholz, M. Raghu, M. Wattenberg, I. Goodfellow, and G. Brain, "The relationship between high-dimensional geometry and adversarial examples," 2018, *arXiv:1801.02774*.
- [165] A. Sarıkaya, B. G. Kılıç, and M. Demirci, "RAIDS: Robust autoencoder-based intrusion detection system model against adversarial attacks," *Comput. Secur.*, vol. 135, Dec. 2023, Art. no. 103483.
- [166] O. F. Tuna, F. O. Catak, and M. T. Eskil, "Closeness and uncertainty aware adversarial examples detection in adversarial machine learning," *Comput. Electr. Eng.*, vol. 101, Jul. 2022, Art. no. 107986.
- [167] Y. Peng, G. Fu, Y. Luo, J. Hu, B. Li, and Q. Yan, "Detecting adversarial examples for network intrusion detection system with GAN," in *Proc. IEEE 11th Int. Conf. Softw. Eng. Service Sci. (ICSESS)*, Oct. 2020, pp. 6–10.
- [168] W. Xu, D. Evans, and Y. Qi, "Feature squeezing: Detecting adversarial examples in deep neural networks," 2017, *arXiv:1704.01155*.



SABRINE ENNAJI received the Engineering degree in cybersecurity from the National School of Applied Sciences, Kenitra, Morocco, in 2019, and the Ph.D. degree in artificial intelligence from the National School of Applied Sciences, Fez, Morocco, in 2023.

Currently, she is engaged in advanced cybersecurity research with the Sapienza University of Rome, Italy, with a specialized focus on adversarial attacks within structured contexts, particularly network traffic. Her research has been presented at several international conferences, significantly contributing to the development and enhancement of intrusion detection systems and AI-driven security frameworks.



FABIO DE GASPARI (Member, IEEE) received the Ph.D. degree in computer science from the Sapienza University of Rome, in 2019. He is an Assistant Professor with the Department of Computer Science, Sapienza University of Rome. In addition to his research, he has experience in industry, having worked as an Intern with the Cisco Systems' Research Center, France, where he focused on the security of information-centric networking. He has contributed to the

field through several publications, with over 664 citations, reflecting his active engagement in research. His research focuses on cybersecurity, with particular interests in security and privacy, active defense, networks, and machine learning. His diverse experience in both academia and industry underscore his commitment to advancing cybersecurity through research and practical applications.



in high-impact journals, reflecting his expertise in the field of secure AI and data protection. His research focuses on cybersecurity, privacy, and deep learning, with particular interests in adversarial machine learning, data poisoning, and secure federated learning.



ALICIA KBIDI received the degree from the ESSEC Business School, focused on innovation and new business models. She is an Innovation Specialist with the Sapienza University of Rome, specializing in marketing, communication, and the development of new business models. In her current role with the Sapienza University's IT Department, she assists researchers in evaluating the business potential and feasibility of their ideas. Additionally, she has been involved in projects,

such as GEN4OLIVE, where she has applied her expertise in marketing and new business models to the agricultural sector.



LUIGI VINCENZO MANCINI received the Ph.D. degree in computer science from the University of Newcastle Upon Tyne, U.K., in 1989. He is a Full Professor with the Sapienza University of Rome. Formerly, he has been the Vice Dean of the Facoltà di Ingegneria dell'Informazione, Informatica e Statistica. He has participated in several national and international research projects in the area of security and privacy, which include SUNFISH Project, funded by the EC Horizon 2020 Program under grant agreement No. 644666; and the Geopolitics Aware INternet Strategies (GAINS) (ref. HOME/2013/CIPS/AG/4000005057) funded by European Commission—Directorate General Home Affairs. He published more than 100 scientific papers in international conferences and journals and received more than 6000 citations. He served on the program committees of several international conferences, such as ACM Conference on Computer and Communication Security, ACM Symposium on Access Control Models and Technology, European Symposium on Research in Computer Security, and Financial Cryptography and Data Security Conference.

...